claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\agent-a4d56eb210a751dad.jsonl`
- SHA-256: `9cd83568ed61f968e4ab029d0fccb50891cb58be6094e19010a2d0a530218ccf`
- Source modified: `2026-07-03T16:40:49+00:00`
- Imported at: `2026-07-05T16:48:26+00:00`
- project: `subagents`
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`

Transcript

1. user (2026-07-03T16:40:06.556Z)

```
You are a code-review finder agent. Review target: PR #451 of the badminton-highlight-indexer
repo, checked out at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer on
branch codex/alpha-readiness-plan (HEAD cd07339). Get the diff with: git diff
origin/master..HEAD. Docs-only PR: new docs/ALPHA_LAUNCH_READINESS.md plus index additions.

ANGLE ? reuse/duplication. Check whether the new tracker duplicates content that already lives
in existing docs in a way that will drift: read docs/NEXT_STEPS.md, docs/ROADMAP.md,
docs/DOC_STATUS.md, docs/COMPUTE_DECOUPLED_SERVING/README.md (its §7 tracker) and compare with
the new doc's §7 deliverables and §3 snapshot. Flag rows/claims that restate another doc's
source-of-truth without pointing at it (creating two sources of truth that will diverge), or an
existing tracker that already covers the same deliverable (name it). Only flag concrete drift
risks where you can name both locations; the doc's stated intent is to CONSOLIDATE pointers, so
mere cross-referencing is fine.

Return up to 6 candidate findings as a JSON array of {file, line, summary, failure_scenario}.
failure_scenario = the concrete maintenance cost. Return [] if none. Your final message must be
only the JSON array.
```

2. assistant (2026-07-03T16:40:08.430Z)

```
I'll analyze PR #451 of the badminton-highlight-indexer repo for reuse/duplication issues in the
new docs. Let me start by getting the diff and reviewing the relevant documentation files.
```

3. user (2026-07-03T16:40:09.382Z)

```
diff --git a/docs/ALPHA_LAUNCH_READINESS.md b/docs/ALPHA_LAUNCH_READINESS.md
new file mode 100644
index 0000000..4b3d49d
--- /dev/null
+++ b/docs/ALPHA_LAUNCH_READINESS.md
@@ -0,0 +1,171 @@
+# Alpha Launch Readiness Plan
+
+> **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**
2026-07-03; **Last updated:** 2026-07-03
+> **Lifecycle:** `DRAFT -> IN PROGRESS -> DONE -> archived`
+> **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed in
`docs/README.md` and `docs/DOC_STATUS.md`.
+> **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`,
`COMPUTE_DECOUPLED_SERVING/`, and the sibling `khelsutra` Studio docs.
+> **Honesty note:** `[verified]` means checked against the local synced repos on 2026-07-03.
`[analysis]` means derived from a local command run on that snapshot. This is not legal,
financial, or launch approval.
+
```

```diff
+---
+
+## 0. TL;DR
+
+The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has 15 badminton videos; `[analysis]` the local manifest needed path rebasing before manifest-driven commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion schema; `[analysis]` the n=15 quality baseline is stale and must be intentionally refreshed before it can gate anything.
+
+The first controlled alpha should be a hosted, authenticated product loop: upload a video, process it, select rallies, compile, and download a reel. The owned-model/commercial rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the WASB checkpoint provenance is resolved.
+
+## 1. Problem & goal
+
+The previous launch blockers were spread across roadmap docs and mixed together three different questions:
+
+- **Can an invited tester use the product without SSH?**
+- **Can the 15-video golden corpus now support honest quality gates?**
+- **Can we claim or ship an owned/commercial rally detector?**
+
+This doc separates those into executable workstreams. The goal is to reach an invite-only alpha where the product is usable end-to-end, the quality claims are honest and reproducible, and the remaining model/legal gates are explicit rather than accidentally implied by launch activity.
+
+## 2. Decisions locked
+
+| # | Decision | Source / date | Implication |
+|---|---|---|---|
+| D1 | **Product alpha and owned-model launch are separate decisions.** | `NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled hosted product loop without claiming the owned model is commercially cleared. |
+| D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling, compile, and download are launch-critical. |
+| D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** | `data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature completeness, baseline refresh, and gate ratification come before quality claims. |
+| D4 | **Do not promote served Gate A by default from the current n=15 evidence.** | `backend.eval.serve_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-windowing safety; keep it as an opt-in/analysis lever. |
+| D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13.** | sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha can use manual/offline corpus promotion; automated flywheel follows after serving and gates are stable. |
+
+## 3. Verified snapshot, 2026-07-03
+
+All five Git checkouts were fast-forwarded before analysis:
+
+| Repo | Branch | Synced HEAD |
+|---|---|---|
+| `badminton-highlight-indexer` | `master` | `e4914d1` |
+| `khelsutra` | `master` | `b0f70a5` |
+| `rally-annotator` | `main` | `ef9e088` |
+| `sports-data-collector` | `master` | `bfc014f` |
+| `sports-obsreport` | `main` | `a0c7252` |
+
+Corpus/eval facts:
+
+- `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says `output/human_lovo_manifest.json` is
```

```
the ingested golden corpus at `n = 15`.
+- `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest found
all 15 trajectories and labels in this workspace.
+- `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
+- `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only 6
videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
+- `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured
proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
+- `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor
eval_baselines/heuristic_lovo_n6.json` produced learned LOVO mean F1 `0.551`, `ship=False`;
blockers remain stale n=6 floor/sign test, uncalibrated ship target, and WASB C3 provenance.
+- `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15> --features-dir
output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off tolF1 `0.1382`,
but marked the frozen baseline stale due to config/corpus changes.
+- `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity group`
currently fails at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`.
+
+## 4. Launch strategy
+
+### 4.1 Alpha lane: make the product usable
+
+The alpha lane is successful when an invited tester can open the app, authenticate, upload a
video, watch the job progress, pick rallies, compile a reel, and download it. This lane should
avoid model-quality promises beyond the measured state and should default to the most reliable
serving path.
+
+### 4.2 Quality lane: make the 15-video corpus gate-ready
+
+The quality lane is successful when the 15-video corpus is portable, reproducible, feature-
complete, and backed by refreshed baselines. Only then should R2 tolF1, Gen-0/TemporalMaxer, or
other model/promotion decisions be ratified.
+
+### 4.3 Flywheel lane: make tester feedback useful
+
+The flywheel lane is successful when user corrections can flow into the collector/vault path
with consent/provenance and a visible re-eval delta. This is valuable for alpha learning, but it
should not block the first controlled alpha if manual promotion can bridge the gap.
+
+## 5. Risks and guardrails
+
+| Risk | Why it matters | Guardrail |
+|---|---|---|
+| Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
+| Stale local manifests create false confidence | Manifest-driven tools skipped or failed until
paths were rebased | Make the corpus manifest portable and add a path/fixture smoke check |
+| 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still n=6
today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
+| Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly reported
stale baselines | Refresh baselines only after reviewer sign-off and record the command/output |
+| P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but can
consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha lane is
already stable |
+| WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained model
regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product launch
tasks |
+
+## 6. Honest limits
+
+Completing this doc's alpha lane does **not** prove the owned model is ready, that WASB weights
are commercially clean, that the product can serve unbounded public traffic, that multi-sport is
```

```
ready, or that every tester correction automatically improves the model.
+
+Completing the quality lane does **not** by itself deploy a service. It only makes the
measurement surface honest enough for a launch decision.
+
+## 7. Deliverables & progress tracker
+
+Legend: TODO = not started, IN PROGRESS = actively being worked, DONE = shipped/verified, GATED
= blocked on owner/legal/platform decision. One PR per row unless the row explicitly names a
cross-repo batch.
+
+| ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
+|----|-------------|--------------|------------|--------|--------|----|
+| A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-indexer`
| - | No | IN PROGRESS | #451 |
+| A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth
ON, upload -> process -> jobs -> compile -> download verified | `khelsutra` + `badminton-
highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
+| A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that
fails loudly when labels or trajectories are missing | `badminton-highlight-indexer` | A0 | No |
TODO | - |
+| A3 | Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile
collector's six-video source manifest with the 15-video eval corpus | `sports-data-collector` +
vault | A2 | Owner upload/storage scope | TODO | - |
+| A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
| Reviewer sign-off | TODO | - |
+| A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6` floor for
future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | TODO | - |
+| A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the
n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |
+| A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | TODO | - |
+| A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |
+| A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-
serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner
approval for M0.4 scope | TODO | - |
+| A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold,
significance rule, and what claims are allowed | Owner + `badminton-highlight-indexer` | A4, A5,
A8 | Owner decision | GATED | - |
+| A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred,
document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance |
GATED | - |
+| A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion,
`train|reeval` job, and delta-F1 surface | `khelsutra` + `badminton-highlight-indexer` +
collector | A1, A3 | Owner D13 reopen | GATED | - |
+| A13 | Alpha launch review: runbook, privacy/consent copy, known limits, rollback/disable
steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner launch decision | TODO | - |
+
+## 8. Concrete next PR sequence
+
+1. **Docs PR:** land this tracker.
+2. **Corpus portability PR:** rebase/portable manifest tooling plus missing-file smoke test.
+3. **Hosted alpha PR/batch:** health/auth/upload/process/job/compile/download verification and
runbook updates.
+4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor refresh.
+5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun
promotion/rejection decisions.
+6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy limits,
and launch checklist.
+
+## 9. Open questions
```

````
+
+1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or should
P10/P11 be reopened before any tester wave?
+2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU, GCE L4
VM, or another approved GPU host?
+3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains unresolved?
+4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
+5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every clip,
or should the eval loaders support mixed schemas with explicit capability tags?
+
+## 10. Definition of done
+
+- [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH.
+- [ ] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
+- [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely on stale
local absolute paths.
+- [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
+- [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
+- [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected on the
15-video corpus.
+- [ ] Ship-gate target and allowed alpha claims are recorded.
+- [ ] WASB C3 status is recorded and reflected in alpha messaging.
+- [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for alpha-
plus.
+- [ ] Owner records GO/NO-GO for the first alpha wave.
+
+## 11. References
+
+Internal docs:
+
+- `docs/NEXT_STEPS.md`
+- `docs/ROADMAP.md`
+- `docs/data_pipeline/GOLDEN_VIDEOS.md`
+- `docs/R2_EVAL_METRIC_DESIGN.md`
+- `docs/RALLY_DETECTION_FIXES_PLAN.md`
+- `docs/COMPUTE_DECOUPLED_SERVING/README.md`
+- sibling `khelsutra/NEXT_STEPS.md`
+- sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
+- sibling `khelsutra/docs/PER_RALLY_SELECTION.md`
+- sibling `sports-data-collector/data/golden_source_videos.json`
+
+Local verification commands used for the 2026-07-03 snapshot:
+
+```powershell
+python -m backend.eval.fusion_compare --features-dir output
+python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
+python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor
eval_baselines\heuristic_lovo_n6.json --version analysis-n15 --out <temp-out>
+python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir output
--no-report
+python -m backend.eval.ablation --features-dir output --granularity group
+```
+
+---
+
+### Changelog
+
+- 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`, golden-corpus
docs, local eval artifacts, and the current 15-video corpus state.
diff --git a/docs/DOC_STATUS.md b/docs/DOC_STATUS.md
index 34411fd..2541672 100644
--- a/docs/DOC_STATUS.md
+++ b/docs/DOC_STATUS.md
````

```
@@ -30,6 +30,7 @@ Before a foundational/tracked doc moves to `docs/archives/`:
 ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the completeness source-of-truth)
 | Doc | Status | Archive-ready? | Next action |
 |---|---|---|---|
+| `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created 2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-model/commercial rally-detection gates | No | Execute §7 in order: corpus portability + n=15 baselines, hosted alpha serving, ablation/fusion reruns, then launch GO/NO-GO |
 | `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M9 LARGELY SHIPPED** (#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights pushed (`weights/0.1.0-l4/`) | No | M11 flywheel + M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap) |
 | `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** (§7) | No | Drives the R-series; complete when §7.1 single quality number is attributed |
 | `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster |
diff --git a/docs/NEXT_STEPS.md b/docs/NEXT_STEPS.md
index f186ff9..a9f35a8 100644
--- a/docs/NEXT_STEPS.md
+++ b/docs/NEXT_STEPS.md
@@ -9,7 +9,7 @@

 ## ? Prioritized pending ? all tracks
 **P0 ? highest leverage / unblocks the most**
-0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service** (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record: [[ui-driven-cloud-serving-gap]].
+0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service** (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record: [[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:** `docs/ALPHA_LAUNCH_READINESS.md` §7.
 0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip install` ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** **(1)** P3 **GPU validation + screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB 0-rally tuned-config blocker), **(2)** P2b-install (uv/script + truly-local default config), **(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.
 1. ? **`COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All §8 gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a **$100/?10k cap** (D17).
 2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus training feed on
```

it.
diff --git a/docs/README.md b/docs/README.md
index 0df1a69..58be5ff 100644
--- a/docs/README.md
+++ b/docs/README.md
@@ -15,6 +15,7 @@
 - DOC_STATUS.md ? **? living Documentation Status & Archival Register**: every foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here; update rows as docs complete.

 ## Product / platform plans
+- ALPHA_LAUNCH_READINESS.md ? **? Active tracked project (`DRAFT`, created 2026-07-03):** cross-repo alpha readiness tracker after the golden corpus reached 15 videos. Separates the hosted product-alpha lane (shareable upload?process?pick?reel) from the owned-model/commercial rally-detection gate (ship target + WASB-C3 still unresolved). §7 is the source of truth for the concrete next PR sequence: corpus portability, n=15 baseline refresh, ablation/fusion reruns, hosted serving, alpha copy/launch review.
 - [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? **? Active tracked project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend` (what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU, in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI). Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY proof for $0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a [runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). §7 M0?M10 tracker is the source of truth.
 - [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED + ARCHIVED (2026-06-21).** Lifted the rally pipeline onto a **rented cloud-native GPU** doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner**; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past_projects/DECOUPLED_COMPUTE/` (incl. `DEPLOY_REPORT_GCP_2026-06-20.md`).
 - PACKAGING_PLAN.md ? **? Active tracked project (`IN PROGRESS`, 11 PRs merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ? `pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA** single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision D10/O1). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP_APP_PLAN.md` narrative.** §7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **? Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

4. assistant (2026-07-03T16:40:12.895Z)

Now let me read the existing documentation files to check for duplication and drift risks:

5. user (2026-07-03T16:40:13.666Z)

1 # Next steps (live) ? refreshed 2026-06-21
2
3 > **How to read:** this top section is the **current cross-track state + prioritized pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed **owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it is now *one of several* active tracks. Live blow-by-blow: memory `current-state.md`. Doc lifecycle/hygiene: `docs/DOC_STATUS.md`.
4
5 ## ? Where we are (2026-06-21)
6 **DECOUPLED_COMPUTE delivered on GCP + archived** (#270); **compute = GCP only** (ADR-013, DO dropped). **`COMPUTE_DECOUPLED_SERVING`** subproject spun up (#272, **ADR-014**) ?

one pure core + local/cloud profiles; **M0?M9 LARGELY SHIPPED** (ADR-014 ratified 2026-06-21;
M1?M4 #279?#283, M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF). **Per-video
workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273).
The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality /
owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.
7
8 > **? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).** The rally-detection product is
not cleared for an external/commercial launch; we stay in **development**. This is the
expected default for the dev phase. Two unblockers gate a future GO: **(1)** set + meet the
ship-gate target (#172) ? currently unset, needs corpus growth (and the R2 tolF1 ablation,
item 8); **(2)** resolve the **WASB shuttle-detector weight provenance** (commercial-clean
licensing). P2b (#396) was a do-no-harm quality increment, **not** a launch gate ? merging dev
PRs ? launching. Re-decide only when both unblockers clear.
9
10 ## ? Prioritized pending ? all tracks
11 **P0 ? highest leverage / unblocks the most**
12 0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few
recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-
only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra
`web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable
service (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)**
the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ?
poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290,
default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs
our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full
record: [[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus
review:** `docs/ALPHA_LAUNCH_READINESS.md` §7.
13 0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable
batteries-included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip
install` ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-
home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades
(P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** **(1)** P3 **GPU
validation + screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ?
native-WASB 0-rally tuned-config blocker), **(2)** P2b-install (uv/script + truly-local default
config), **(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then
P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` /
`[[next-session-packaging-app]]`.
14 1. ? **`COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014
ratified). M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged
(default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All §8
gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel
ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12
real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a
$100/?10k cap (D17).
15 2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3
matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus
training feed on it.
16 3. **[owner + Claude · GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the
`0.1.0-l4` bundle is n=2 plumbing-only).
17
18 **P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)**
19 3b. **[Claude] Five-repo audit health remediation ? ? COMPLETE (2026-07-02).** All waves
(P0?P28) landed and the last residuals were closed or filed as issues; the umbrella tracker was
archived to `docs/archives/past_projects/AUDIT_REPORT_khelsutra-
guru_2026-07-02-COMPLETED-2026-07-02.md`. Remaining owner-gated/future work lives as GitHub
issues: #443 (CORS default), #444 (corpus Phase-2 upload), #445 (GCS eval plumbing), plus
#301/#327/#332/#384.
20 4. **[Claude] `COMPUTE_DECOUPLED_SERVING` batch ? REMAINING: M11 + M12**
(M5/M6/M8/M9-auth ? merged this session). **M11** = `backend/flywheel.py` capture?store?shadow-
eval?goldenize plumbing reusing `workspace.py::for_video` + `backend/eval/experiment.compare` +
`backend/eval/promotion.py::promote_if_served_safe`, behind `flywheel.*`=OFF + a faked
`consent_attested` (? privacy-sensitive ? give it fresh focus). **M12** =
`backend/storage/gdrive_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme
(regex already allows the hyphen; add to `_KNOWN` + `_backend_class` + the StorageConfig
Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://`

untouched. ? **M9-consent cols = a new v7 migration** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.
21 5. **[Claude] Per-video workspace P1?P6** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of `detector/`+`tmp/` *[the `output/chunks_*` de-hardcode base = ? done via #282]* ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.
22 6. ? **Ops-Agent codified (#291)** ? `deploy/gcp/create_gpu_vm.sh` (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])
23 7. **[owner] Set the ship-gate target (#172)** ? needs corpus growth.
24 8. **[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate** (the ablation) ? needs GPU-extracted golden features.
25
26 **P2 ? later / gated / deferred**
27 9. Rally-quality **P3/P4** (person-fusion LOVO, audio cue) ? needs GPU golden features.
28 9b. ? **DONE (#396):** the R4 density fix (finding B / P2b in `RALLY_DETECTION_FIXES_PLAN.md`) is **promoted to default-ON**. Golden-set A/B at the **SERVED** windowing showed tolF1 0.353?0.364 (?tolF1 +0.010, no regression), so `inactivity_temporal_density` default was flipped to True. (Landed flag-gated default-OFF in #394.) Fixed the sparse-tracking quirk where one fast sample after a detection gap held the rally tail open.
29 10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).
30 11. **Multisport** (table-tennis first, then pickleball).
31 12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within `COMPUTE_DECOUPLED_SERVING`; owner/legal-gated).
32 13. **WASB-C3** weight provenance (gates *sharing* a trained model) ? legal.
33 14. **Rename repo off "badminton"** (owner, BACKLOG).
34 15. Owner housekeeping: retire the shared **DO droplet** + **DO token**; collector **`md5`** export (collector repo).
35 16. **PMF field validation** (owner-side, business track).
36
37 **Doc-hygiene (tracked in `docs/DOC_STATUS.md` §4):** ? #1 per-video reconcile + ? #2 this refresh; **pending:** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary confirm); confirm the COMPLETED top-level stubs; verify `HOSTING_PLAN`/`UI_PROPOSAL` currency.
38
39 ---
40
41 ## Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)
42
43 **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness design docs merged,
44 owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
45 ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see
46 `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**; **PR-2
47 upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload MD5 byte-identity +
48 compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep #90?#98 landed** ? ruff/mypy/
49 coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into
50 `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-verified exact).
51 **Authoritative roadmap:**
52 `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-blow:** memory
53 `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next," refreshed each session.
54
55 ## ? Next-session / GPU-box pickup (2026-06-15)
56 **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**), train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16 drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already

present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 / bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level, not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing ? Actions minutes / spending limit).

Immediate next step ? the real P3 LOVO measurement (this box has a GPU + `torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden features + `.rallies.csv` labels are not yet here):

1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"` ? produces `*_golden_features.json` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow: **[`GOLDEN_DATA_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir <shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir <shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).

Also queued (owner-gated / pending input):
- Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
- Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent will do it).
- Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config bug.

Leaving (bulky / M2 ? no start without owner go): ActionFormer (M0.4), event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multisport, repo rename, PMF execution.

Where we are (one paragraph)
ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (`training/`) behind a CI-enforced import wall; the featurizer is single-sourced (`backend/features/rally_seq.py`, train==serve); the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest output/human_lovo_manifest.json --floor eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0 (LOVO **0.626**, floor passed, sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned corpus is in the collector** (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The **Gemini battery is done** (see table) ? the moat is quantified.

The quality table that now drives strategy (IoU 0.5 vs human golden labels)
Path	Mahadevpura (clean-ish)	Kushagra (multi-court)
Heuristic (velocity windowing)	0.657	0.157
Gen-0 owned head (LOVO, n=6)	0.744	0.561
Two-stage WASB+Gemini refine	0.83	?
Gemini wrapper (`gemini-flash-latest`)	**0.93**	**0.66**

Reading: clean footage is commoditized (serve it with Gemini for cents). **Multi-court drops the commodity wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game pickups + dead-time merges, the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes (uploads candidate clips only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry load-bearing.

92 Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).
93
94 ## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in progress)
95 **This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR #112 + session). All implementation is **owner-gated** (per explicit in the plan); each step = ONE PR off `master`.
96
97 **Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs #156/#157/#160/#161/#163 + #165 + final records).** See archived `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every pre/post/review) and `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive done; clean slate.
98
99 - **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion layer) ·
100 `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) ·
101 `docs/ROORA_LABELING_GUIDELINES.md` (v8) · `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model).
102 - **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a rally" bug is **NOT a missing capability** ? `backend/pipeline/detectors/rally_gate.py` already implements the net-crossing **"Gate A"** (`apply_gate(..., min_crossings=2)`) that kills service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no `min_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via fusion_hybrid) was the single cheapest, highest-confidence quality win.** (P2 FusionSegmenter specifics: registry `fusion_hybrid`, **default-OFF**, seeds from substrate, persists `net_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `tests/test_served_gate_a_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)
103 - **Current status (2026-06-14):** P1 (person cue extraction to reusable `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via hit_detector.
104 - **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):**
105 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)
106 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant (players_in_court + two_sided + colocation etc. from `fusion_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).
107 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.
108 - **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default OFF**; promote to default **only on measured LOVO lift** through `run_regression.py` (exit 0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
109 - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden features + `.rallies.csv` labels, which are not yet present in this checkout.
110 - **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are exhausted; full details + citations in [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) §5.6.** An owner-commissioned 31-reference review of the quality report *validated* the methodology and surfaced these **future** levers (research inputs, not active work, not new claims; verify citations before any submission):
111 1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as *"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."*
112 2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TenniSet/T-DEED) ? turns R2 into a defensible publishable contribution.
113 3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).

114 4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) ? see "M0.4 next increment" below.
115 5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator noise.
116 6. **Name the eval?serve lesson "pipeline distribution skew"** ? a citable paper "lesson."
117
118 **Discipline reminder:** owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.
119
120 ## Prioritized next steps
121 1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.
122 Priority order for new videos: **(a) multi-court badminton** (the target distribution AND where the ship target
123 lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1 0.909; pickleball labels are
124 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per
125 `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool** (consent guardrail). Per video:
126 label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ? re-run the Gen-0 harness.
127 The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is reachable this month.
128 2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court
129 reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93
130 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with
131 paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
132 3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the
133 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the
134 LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer TemporalMaxer** (parameter-free
135 max-pool, ~2.8× fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review
136 (`RALLY_QUALITY_RESEARCH.md` §5.6) ? better fit for the cheap/local/fast mandate.
137 4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`); merged-prototype LOVO across
138 badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated
139 by C3 (provenance multiplies per sport).
140 5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:
141 "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies
142 (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:
143 `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner ratified the
144 physical-visit approach; reframed product-feedback-first per owner spec):**
145 `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact placeholders and recruits) +
146 `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student script, demo-after-Q6
147 rule, ground rules, Phase-2 record-and-show-back stretch goal) + `docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
148 capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md` (G/A/R anchors + **pre-registered
149 decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first

```
visit).
6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified working; confirm old-key
   deletion in AI Studio); **rename repo off "badminton"** (owner action,
   in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the battery completed.

## Pointers / artifacts
- Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008 (`docs/archives/decisions/DECISIONS.md`) ·
  Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
  `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
- Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact bundle; weights gitignored,
  manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
- Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest `output/human_lovo_manifest.json`
  (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\` ·
  labels: `...\Collect\Golden Labelled\`.
- Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
  (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a cold/prepaid project burst-throttles
  with a MISLEADING "credits depleted" 429; ramp gradually.
- ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next milestone. Committed next
  PLATFORM slice = async job model (single-tenant).

```

6. user (2026-07-03T16:40:13.822Z)

```
# Roadmap ? from annotated data to a self-improving, offline rally segmenter

This document is the **narrative thread**: how the indexer evolves from an
AI-API-dependent prototype into a self-contained tool that learns rally detection
from its own commercially-licensed, annotated video corpus. It explains *how we
got here*, *why each step matters*, and *how it shapes the final tool*. Decisions
are recorded as ADRs in [DECISIONS.md](archives/decisions/DECISIONS.md); the scoring mechanics live
in [EVALUATION.md](EVALUATION.md).

> ? **Phase 4 update:** the authoritative owned-model path is now
> [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md) ? a learned **per-frame TAL head**
> (ActionFormer/ASFormer over WASB trajectory features), not the candidate-window classifier sketched below.
> **Audio is parked, not "gated on golden labels":** the E1 probe and a follow-up foreground-audio test both came back
> weak (~0.6 AUC), so it is de-prioritized even though labels now exist (see the plan's Gen-1 note).

## The thesis

The indexer detects badminton rallies in long recordings. Its strongest mode
(`yolo_hybrid`) is a two-stage pipeline: cheap **permissive person-detection
player-motion** filtering (Stage 1, local ? torchvision Faster R-CNN + ByteTrack
tracking) followed by **Gemini** semantic refinement (Stage 2, paid API). That
works, but it costs money, needs the network, and never improves on its own.

> *Naming note:* the `yolo_hybrid` registry key is retained for back-compat, but
> AGPL YOLOv8 was removed and replaced with a BSD/MIT-licensed torchvision +
> ByteTrack stack ? see **ADR-005**. "YOLO" in this doc means the local
> player-motion gating stage, not the Ultralytics library.

A sibling project, **sports-data-collector**, curates sports videos with
```

per-rally `[start, end)` annotations and a commercial-license flag. The thesis of this roadmap: **use that annotated, commercially-usable data to (1) measure the pipeline objectively and (2) train an offline replacement for the paid Gemini stage** ? turning the tool into one that improves as more data is curated, runs air-gapped, and generalizes across sports.

Why it matters: it removes the per-run API cost and network dependency, gives us a defensible quality metric instead of eyeballing, and creates a flywheel ? more curated matches ? a better offline segmenter ? better highlights.

The four phases

Phase 1 ? Bridge the repos (DONE)
A sport-agnostic `curate export` in the collector emits the indexer's `start,end` ground-truth CSVs plus a `manifest.json` pairing each CSV with its video, gated on the `is_commercial` flag. ? *Why:* one clean producer/consumer contract that works for all five supported sports (badminton/tennis/cricket/pickleball/table-tennis) alike. See **ADR-002**.

Phase 2 ? Acquire full-resolution video (DONE)
The corpus was 256×144 (the original crawl used `worstvideo`). The `fetch` command re-acquires every video at **best available resolution (1080p here)**, **in place** (preserving annotations), via **authenticated Firefox Premium cookies** and yt-dlp's `default` client. ? *Why:* YOLO/shuttle detection needs real pixels; and the journey surfaced non-obvious traps (SABR streaming ? 144p; Chrome DPAPI; rate limiting) now encoded in the tooling. **22/22 fetchable matches are now 1080p.** See **ADR-003**.

Phase 3 ? Evaluate + tune (IN PROGRESS)
`scripts/run_phase3_eval.py` walks the manifest, processes each video, and scores detected rallies against the ground truth ? per video and corpus-aggregate (precision/recall/F1, mean IoU, boundary error), for both the `candidates` (raw detector) and `final` (refined) stages. ? *Why:* the candidates-vs-final split tells us *where* error lives ? a **detection** problem (candidates miss rallies) vs. a **segmentation** problem (candidates catch them, refinement is loose). That diagnosis decides where effort goes, and establishes the baseline any trained model must beat.

Phase 4 ? Learn an offline, API-free segmenter (NEXT)
Replace the paid Gemini refinement with a **lightweight classifier trained on the rally annotations**, consuming features the pipeline already computes and stores (`candidate_segments.stats`). No external API, no new labeling. ? *Why:* this is the payoff ? a free, offline, self-improving rally mode. The **substrate was chosen from data**: the ADR-004 update (2026-06-02) measured the YOLO/player-motion candidate stage at **recall ~0.05 @ IoU 0.5** on `shuttleset_8` (median best coverage of a rally only ~20%) ? too low a ceiling to build on. So the offline segmenter's substrate is **WASB shuttle-motion** (ADR-001), not player-motion. See **ADR-004**.

Complementary local cues (beyond the segmenter swap)

Once the substrate is WASB shuttle-motion, the next lever is **fusing additional fully-local cues** to raise rally recall/precision without reintroducing a paid API. The first one explored is **audio "thwack" hit-detection** (**ADR-007**, accepted as a direction 2026-06-07): a pure-numpy spectral-flux onset detector in `backend/pipeline/audio/`. The **E1 GO/NO-GO probe** has run on real GoPro audio ? verdict **AMBER / leaning-negative**: recall signal is real (hit-clusters recover WASB-missed rallies) but the classical-onset precision ceiling (~2.2×) is too weak to fuse cleanly, and a band-pass doesn't rescue it. The only path past that is a *learned timbre classifier* ? but per the owned-model plan audio is now **parked** (the foreground-audio idea also failed), **not** revived now that labels exist. See `archives/research/AUDIO_E1_FINDINGS.md`. Pose/court serve-gating is the more credible #2 cue.

The recurring blocker across Phases 3?4 and the audio work is the same: **more golden labels** is the single highest-leverage move (see the cross-cut note below).

How this shapes the final tool

- **Offline & free by default.** The production rally mode becomes a local model;
  Gemini is kept only as an upper-bound reference, not a dependency.
- **Self-improving.** Every newly curated, annotated match is more training signal;
  re-training is a local, deterministic step.
- **Measurable.** Every change is scored against the golden set, so "better" is a
  number, not a vibe.
- **Sport-generic.** The label ? feature ? classifier machinery and the export
  bridge already span sports; adding a sport is data, not new architecture.
- **Commercially clean.** Only `is_commercial` data flows through the bridge, so the
  trained model and any derived product rest on a licensable foundation.

Status at a glance

Phase	What	Status
1	Export bridge (`curate export` ? manifest + CSVs)	? Done
2	1080p acquisition (`fetch`, in-place, cookies)	? Done (22/22)
3	Eval + tuning loop (`scripts/run_phase3_eval.py`)	? In progress
4	Offline learned segmenter (WASB shuttle-motion substrate)	? Next (substrate decided: WASB, ADR-004)
+	Audio hit-detection cue (`backend/pipeline/audio/`, ADR-007)	? E1 probe run ? AMBER (~0.6 AUC), parked / de-prioritized

> **Cross-cutting:** golden-set ground truth (which powers Phases 3?4 and on-demand regression testing)
> is produced via a vendor process + pluggable annotation provider ? see
> [GOLDEN_SET_VENDORING.md](archives/roora-vendor/GOLDEN_SET_VENDORING.md) and **ADR-006**. As of 2026-06-02 the annotation
> guideline is authored (collector `docs/annotation/`) and **Roora** (Bengaluru) is selected for an
> initial **3-sport pilot** (badminton, pickleball, table tennis). Input-capture quality requirements
> that bound achievable precision/recall are tracked in [VIDEO_RECORDING_GUIDELINES.md](data_pipeline/VIDEO_RECORDING_GUIDELINES.md).
>
> **Observability:** every processed video now emits an always-on per-video diagnostics
> report (spec-driven warnings/errors with FAQ refs, soft-dependency on `obsreport`), so
> pipeline regressions and footguns surface without re-running eval ? see
> CODE_MAP.md §2 (reporting/observability).

7. user (2026-07-03T16:40:14.361Z)

Documentation Status & Archival Register

> **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-diligence**. **Last full review:** 2026-07-02.
> **Update cadence:** whenever a doc changes lifecycle state (created / shipped a milestone / delivered / archived), and re-review at each clean-slate.
> **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update it as work finishes** to prove we're done, and (c) **never archive a foundational doc without honestly updating it first** (owner directive, 2026-06-21).

---

0. How to use this
- **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention. §3 is the full register.
- **Marking something complete?** Update the doc's own **§7 progress tracker** (the source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2 archival checklist.
- **This register does NOT duplicate completeness detail** ? it indexes lifecycle +

archive-readiness and points to each tracked doc's §7.
13
14 ## 1. Lifecycle legend
15 `DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal record of a run/launch), **CONVENTION** (a standing rule/guide).
16
17 ## 2. Archival due-diligence checklist ? *the repeatable process (do NOT just `git mv`)*
18 Before a foundational/tracked doc moves to `docs/archives/`:
19 1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts `[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it first.)
20 2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped, where (PRs, paths, artifacts), and what was **deferred / carried forward**.
21 3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`, `docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
22 4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the evolution stays traceable.
23 5. **`git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this in the doc's header).
24 6. **Record the move** ? flip the row here to §3e + a line in memory `current-state`.
25
26 > **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5 delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward WASB-C3/DPA/`md5`), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014 lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.
27
28 ## 3. The register
29
30 ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the completeness source-of-truth)
31 | Doc | Status | Archive-ready? | Next action |
32 |---|---|---|---|
33 | `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created 2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-model/commercial rally-detection gates | No | Execute §7 in order: corpus portability + n=15 baselines, hosted alpha serving, ablation/fusion reruns, then launch GO/NO-GO |
34 | `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M9 LARGELY SHIPPED** (#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights pushed (`weights/0.1.0-l4/`) | No | M11 flywheel + M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap) |
35 | `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** (§7) | No | Drives the R-series; complete when §7.1 single quality number is attributed |
36 | `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster |
37 | `R1_MILESTONE_LAUNCH_REPORT.md` | LAUNCHED (#204) ? terminal record | Soon (with the R-cluster) | Keep with R-series for narrative; archive when the track wraps |
38 | `RALLY_DETECTION_GAP_CLOSURE.md` | DRAFT (owner-gated) | No | G1?G3 levers; part of the R/quality cluster |
39 | `RALLY_DETECTION_FIXES_PLAN.md` | **DONE** ? findings A/B/C resolved; P2b promoted `inactivity_temporal_density` default-ON (#396) | With the R-cluster (deferred per its header; gate = R2 tolF1 ablation + §7.1 number) | Archive with the R-series when the track wraps |
40 | `RALLY_DETECTION_CODE_REVIEW.md` | Terminal snapshot ? findings A (#394), B (#394 ? promoted #396), C (`sdc#50`) all resolved | With the R-cluster | Archive with the R-series |
41 | `PER_VIDEO_WORKSPACE.md` | DRAFT, IN PROGRESS (P0 = #264) | No | ? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once) |
42 | `MULTI_SIGNAL_FUSION_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion, audio) ? needs GPU golden features |
43 | `EXPERIMENT_HARNESS.md` | P1 IMPLEMENTED | No | A/B + promotion gate; supports the quality track |
44 | `PROMINENT_COURT_DETECTION.md` | DRAFT (owner-gated) | No | C-series; multi-track

selection design |
`EXECUTION_POLICY.md`	P1 SHIPPED; P3b next	No	Async job policy; reused by Cloud Run dispatch
`GOLDEN_REGRESSION_FIXTURES.md`	IN PROGRESS (owner-gated)	No	Golden eval fixtures
Data layer ? `docs/data_pipeline/` (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS` · `VIDEO_RECORDING_GUIDELINES` · `GOLDEN_DATA_SHARING`)	LIVING (data, **no-ML**)	n/a	Relocated 2026-06-21 to the isolated data pillar; tracked via `data_pipeline/README.md`. Roora *vendor* docs archived ? `archives/roora-vendor/`.
`archives/past_projects/AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md` **(umbrella)**	**? COMPLETE / ARCHIVED (2026-07-02)** ? the five-repo code audit + folded-in *Remediation tracker* (P0?P28 all closed) + absorbed prior audits. All non-owner-gated residuals landed; owner-gated/future items filed as issues (#443/#444/#445, + #301/#327/#332/#384).	Archived	Owner confirmed archival 2026-07-02; moved to `archives/past_projects/` per §2. Residuals now live as GitHub issues, not in this doc.
`DECISION_SNAPSHOT_REGRESSION.md`	**DRAFT ? decisions LOCKED** (D1?D11 locked 2026-06-25, #383). Zero code shipped for P1?P7.	No	Start P1+P2 (schema + stitch-plan factor); de-risk the P10 main.py split
`DESKTOP_APP_PLAN.md`	DRAFT (owner-gated, nothing built)	No	P0 compute-feasibility spike; relates to truly-local profile
`OWNED_MODEL_IMPLEMENTATION_PLAN.md`	PLAN/roadmap (owner-gated)	No (authoritative roadmap)	Gen-0?Gen-2; the owned-model track
`PERSONALIZATION_PLAN.md`	DESIGN ? owner-adopted; Phase-0	No	Per-user models (DB v5 store landed)
`UI_ALPHA_EXECUTION_PLAN.md`	ACTIVE	No	khelsutra UI alpha track
`I18N_PLAN.md`	Design (P0); **in implementation** (adapted)	No	Executed in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-gated
`POST_166_...RISK_ASSESSMENT.md`	Plan for owner review	No	Milestones/risk doc

3b. Foundational / strategy references (LIVING ? archive only if **superseded**, not when "done")
Doc	Status	Note
`PLATFORM_ARCHITECTURE.md`	Strategy/research	The `ComputeBackend`/`StorageBackend` registries ? realized by COMPUTE_DECOUPLED_SERVING
`COMMERCIALIZATION.md`	? LIVING	Strategy; carries WASB-C3
`VIDEO_LOCALITY_MODEL.md`	THINKING DOC	L0 fully-local = the truly-local profile
`HOSTING_PLAN.md`	PROPOSAL	**? partly overtaken by ADR-013 + the shipped site ? verify (finding #5)**
`UI_PROPOSAL.md`	PROPOSAL (partly shipped)	UX half **in implementation** ? `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc)
`STORAGE_SHARING_MODEL.md`	DRAFT (co-design pending)	?
`ANALYZERS_AND_RECONCILERS.md`	DESIGN (owner-gated)	?
`ROADMAP.md` / `CODE_MAP.md` / `HOW_RALLY_DETECTION_WORKS.md` / `KHELSUTRA_PRODUCT_OVERVIEW.md`	LIVING references	Keep current; not archive targets

3c. Reports & terminal records (archive candidates once their track wraps)
Doc	Status	Note
`REAL_FOOTAGE_VALIDATION_REPORT.md`	validation complete	Motivates R2; archive with the R-cluster
`RALLY_DETECTION_QUALITY_REPORT.md`	report	Quality snapshot; part of R-cluster
`QUALITY_ITERATIONS.md`	LIVING empirical log	**Keep** ? the paper-backbone ablation log ([[paper-coauthor-aim]])
`CODE_AUDIT_AND_TEST_HARDENING` · `HARDENING_LOOP_HANDOFF` · `DEFERRED_HARDENING_PLAN`	COMPLETED ? **finish-archived 2026-07-02** (top-level stubs removed; absorbed into the umbrella's *Prior audits*)	? Full copies under `docs/archives/*-COMPLETED-2026-06-14.md`; inbound links repointed (finding #4 resolved)

3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)
`PROJECT_DOC_TEMPLATE.md` · `NEXT_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` · `EVALUATION.md` · `ABLATION_LAYER.md` · `SETUP_NEW_MACHINE.md` · **`CREATING_REELS.md`** (? end-

user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) · **`CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`** (? operator runbook ? `gs://` videos ? reels on L4 Cloud Run; verified e2e 2026-06-23) · `TRACKNET_WSL_SETUP.md` · `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*
79
80 ### 3e. Already archived (`docs/archives/`)
81 `past_projects/DECOUPLED_COMPUTE/` (? 2026-06-21, #270) · `past_projects/low-regret-moves/` · **`field-pmf/`** (field-ops + customer-feedback, 2026-06-21) · **`roora-vendor/`** (annotation-vendor engagement, 2026-06-21) · `PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md` · `CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md` · `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` · `COMMERCIAL_READINESS_REVIEW.md` · `MONETIZATION_AUDIT.md` · `TEST_RUN_SUMMARY.md` · `research/` · `checkpoints/` · `quality-iterations/` · `past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md` (absorbed ? umbrella) · `past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md` (folded ? umbrella).
82
83 ### 3f. ADR log ? `docs/archives/decisions/DECISIONS.md`
84 **ADR-001 ? ADR-014** (14, all ratified). The compute/serving lineage: **ADR-005 ? 008 ? 009 ? 010 ? 011 ? 012 ? 013 ? 014**. ADR-014 was **RATIFIED 2026-06-21** (owner M0 sign-off for COMPUTE_DECOUPLED_SERVING).
85
86 ## 4. Open due-diligence findings / actions (2026-06-21 review)
87
88 > **? Doc-restructuring done (2026-06-21, owner-directed):** field-ops/customer-feedback ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML **data layer** carved into `docs/data_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data_pipeline` one sets the "no segmentation/stitching" boundary).
89 1. **? RESOLVED (2026-06-21) ? `PER_VIDEO_WORKSPACE.md` supersedes `PER_VIDEO_OUTPUT_LAYOUT.md`.** Folded forward the `video_slug` naming option (workspace §8.5) + the `output/GX010125_run/` manual-prototype precedent; predecessor archived ? `docs/archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE_MAP.md` repointed.
90 2. **? RESOLVED (2026-06-21) ? `NEXT_STEPS.md` refreshed** with a current 2026-06-21 state + a **cross-track prioritized pending list**; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.
91 3. **R-series / quality cluster is ACTIVE** (`R1`,`R2`,`RALLY_DETECTION_GAP_CLOSURE`,`RALLY_QUALITY_RESEARCH`,`QUALITY_ITERATIONS`,`REAL_FOOTAGE_VALIDATION_REPORT`,`RALLY_DETECTION_QUALITY_REPORT`). **Do NOT archive yet** ? archive as a *cluster* once R2 is promoted-or-rejected as the gate + §7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)
92 4. **? RESOLVED (2026-07-02) ? finish-archived the hardening stubs.** The three top-level pointer-stubs (`CODE_AUDIT_AND_TEST_HARDENING`, `HARDENING_LOOP_HANDOFF`, `DEFERRED_HARDENING_PLAN`) were removed; full archived copies remain under `docs/archives/`, inbound links (README, PLATFORM_ARCHITECTURE, DESKTOP_APP_PLAN) were repointed, and the sweep is summarized in the umbrella's *Prior audits (absorbed)*.
93 5. **`HOSTING_PLAN.md` / `UI_PROPOSAL.md` are PROPOSAL but partly overtaken** by ADR-013 (DO dropped) + the shipped khelsutra site. **? `UI_PROPOSAL.md` banner added (2026-06-21)** ? its UX half is now tracked in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; the `I18N_PLAN.md` localization track is likewise in implementation there. **`HOSTING_PLAN.md` banner still pending** (ADR-013 already noted in it).
94 6. **? RESOLVED (2026-06-21) ? `COMPUTE_DECOUPLED_SERVING` M0 approved + M1?M4 SHIPPED** (#279/#280/#282/#283); all §8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized (owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17 $100/?10k cap).
95
96 ## 5. Maintenance
97 Update a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at each clean-slate. This register + the §2 checklist are the operational form of the `CLAUDE.md` archive policy and [[working-agreement-tracked-project-docs]].
98

8. user (2026-07-03T16:40:15.103Z)

1 # Compute-Decoupled Rally Serving ? one pure core, profile-selected compute + storage
2

> **Status:** ? **M0 APPROVED** (owner, 2026-06-21) ? **IN PROGRESS (build M1+)** · **Owner:** Avi · **Created:** 2026-06-21 · **Last updated:** 2026-06-21
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/past_projects/` when DONE)
> **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory `current-state` + `docs/NEXT_STEPS.md` once work starts.
> **Relation to existing docs:** the productionization successor to the archived `DECOUPLED_COMPUTE` (M3/M4); realizes the `ComputeBackend`/`StorageBackend` registries from [PLATFORM_ARCHITECTURE.md](../PLATFORM_ARCHITECTURE.md) §3; the **truly-local** profile is the L0 tier of [VIDEO_LOCALITY_MODEL.md](../VIDEO_LOCALITY_MODEL.md) and the `LocalDesktop` backend of [DESKTOP_APP_PLAN.md](../DESKTOP_APP_PLAN.md).
> **Naming:** this was the "Cloud Run + GPU serving subproject" (ADR-012). Renamed **compute-decoupled serving** because **truly-local is a co-equal, first-class profile**, not an afterthought.
> **Honesty note:** claims tagged `[verified]` (checked against code by the design workflow `wf_56274ff0`), `[design]` (proposed), `[researched]`.

> **? Northstar (D14):** a **mobile, app-like** flow ? point at a **Google Drive** video ? **fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive, next to the source**. Two design promises around it: a user can **switch local?cloud anytime** (D13, a headline feature), and execution is **never a surprise** ? auto-detect *suggests*, the user *confirms* (D15).

0. TL;DR
The rally engine is a **pure function of `(input bytes + config) ? outputs`** in three deterministic stages ? **DETECT ? WINDOW ? STITCH** ? that **no deployment profile may branch on**. A **`ComputeBackend` (deployment profile)** decides *where* the core runs (a local process vs a Cloud Run GPU job); a **`StorageBackend`** decides *where bytes come from* (local FS / USB / mounted Google Drive / bucket). **One startup config** picks both. We ship **truly-local** (local GPU + in-process stitch, zero cloud) and **cloud-serving** (upload `<2GB` / gdrive / bucket ? a Cloud Run GPU job runs detect **and** stitch, so stitch's compute is metered), plus **cpu-mock** for CI. The single most important caveat: detection is **byte-identical only on the same GPU** (cross-GPU is sub-pixel-different, per the 2026-06-20 deploy report) ? so the **parity guarantee is enforced at the WINDOW + STITCH layer**, not detector-CSV bytes.

1. Problem & goal
- **Why now:** DECOUPLED_COMPUTE proved compute *portability* (M0?M2 + M3.5 on a GCP L4) and is archived; its deferred **M3 (serving endpoint + per-video billing) + M4 (scale-to-zero + cost guard)** become this project. The owner needs **both** a hosted service *and* a truly-local mode, with the **core (detect+stitch) unaffected** by which one runs.
- **The two user-facing outcomes** (owner, 2026-06-21):
  - **(a) Hosted service** ? operate on small **local uploads (`<2GB`)** *and* load from **gdrive / a bucket**, spinning up **Cloud Run for both rally segmentation AND stitching** (stitching is compute-intensive ? run it where its cost is accounted for).
  - **(b) Truly-local** ? on a local machine with videos on **local drives/USB + a local GPU**, run inference **and** stitching **completely locally**.
- **"Good" looks like:** the same `video ? rallies ? reel` core gives identical windows + stitched ranges whether it runs on a laptop or Cloud Run; switching is a **config/startup choice**, never a code branch in the core.

2. Decisions locked

#	Decision	Source	Implication
D1	The core is a **pure 3-stage function** (DETECT/WINDOW/STITCH); **no profile branch inside it**.	design `[verified]`	All profile difference lives in config + the two registries.
D2	**Two registries:** `ComputeBackend` (where) selected by `deployment.profile` + `compute_target`; `StorageBackend` (what-bytes) by `input_backend`/`storage.backend`.	ADR-014	Realizes PLATFORM_ARCHITECTURE's `local|hosted|byo_worker`.
D3	**Profiles shipped:** `truly-local`, `cloud-serving`, `cpu-mock` (CI); `byo_worker` reserved/deferred.	ADR-014	Enum/seam reserved so BYO isn't precluded.
D4	In **cloud-serving, STITCH runs on Cloud Run** (co-located with detect) so		

CPU/wall + egress are metered into the per-job cost ledger. | owner (a) | Stitch-on-laptop would hide real cost. |
32 | D5 | **Parity enforced at WINDOW+STITCH** (byte-exact for a fixed trajectory); **detect** is byte-exact only same-GPU. | deploy report 2026-06-20 | Cross-GPU asserted at the rally/window level, never CSV bytes. |
33 | D6 | **Default-OFF, smallest-safe-first, one PR per milestone**; any core-caller change (e.g. configurable output base) ships behind a flag + a Launch Report. | [[launch-report-convention]] | Zero-regression discipline. |
34 | D7 | **Cloud Run cold-start: SCALE-TO-ZERO** ? accept the ~1?3 min cold-start (it's an async job: upload?poll?reel, amortized over the multi-minute job). **No warm pool** (that ~$500/mo idle would break per-video billing). | owner 2026-06-21 | M6/M7. Revisit a warm lane only on a UX complaint. |
35 | D8 | **Pricebook = a versioned DB table**; cost computed in **USD** (matches GCP billing = one source of truth), **displayed in INR** at a configurable FX rate (Bangalore market); re-versioned when GCP prices change. | owner 2026-06-21 | M8. |
36 | D9 | **Edge auth = Cloudflare Access** (reuse the site's existing CF). Lock the Cloud Run **ingress to the CF proxy** + **verify the `Cf-Access` JWT signature** (so a direct hit to the Cloud Run URL can't spoof the identity header). | owner 2026-06-21 | M9. One identity system. |
37 | D10 | **Free-tier "data-for-reels" trade:** free reels in exchange for **training rights** (uploaded footage + derived trajectories/labels); **paid tier = no-train** (privacy is the paid feature). **Carve-outs:** train ONLY on attested **ADULT** footage (minor footage ? reel yes, training pool **NO** ? DPDP/GDPR need verifiable parental consent); train on **DERIVED** data + **auto-delete raw**; **ToS counsel-reviewed**; live training-on-uploads **gated to M9** (public signup). Truly-local needs no DPA. | owner 2026-06-21 | M9 + D12; ties [[paper-coauthor-aim]] / PERSONALIZATION flywheel. |
38 | D11 | **Quality floor: Gemini handoff ON** for cloud-serving until the owned model clears the #172 ship-gate (e.g. ?0.70 multi-court) ? then flip via a **Launch Report**; the owned model runs in **shadow/eval** meanwhile. (Gemini = serving/inference ONLY, never a training source ? CLAUDE.md guardrail.) | owner 2026-06-21 | M7 + D12. |
39 | D12 | **Data-flywheel harness is IN SCOPE (owner add, Q5):** capture the (consented, adult) uploaded video + the Gemini reel/rally output ? **reliably store** in the per-video workspace/bucket ? **run shadow evals** (owned-vs-Gemini, dual-eval-set) ? **promote good ones to the golden TRAINING set** (human-reviewed labels) so the owned model climbs toward the D11 floor (then Gemini flips off). | owner 2026-06-21 | new **M11**; ties D10 (consent) + `data_pipeline/` + the eval/experiment harness. |
40 | D13 | **Profile is user-switchable ANYTIME ? a headline FEATURE.** The same user runs **local today, cloud tomorrow, back to local** ? it's just a config/startup choice; the pure core gives **equivalent** output either way. **Advertise it** ("your machine *or* our cloud ? your call, switch anytime"). | owner 2026-06-21 | Honest caveat: equivalent at the **rally/window** level, not bit-identical across GPUs (D5); a single job runs in one profile, switching is *between* jobs. |
41 | D14 | **NORTHSTAR ? operate toward this:** a **mobile, app-like** flow ? point at a **Google Drive** video ? **fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive, next to the source**. | owner 2026-06-21 | Implies a **`gdrive-api` (OAuth) StorageBackend** (read source + write reel back) ? distinct from the local **mount** backend; **resolves §8 #7**, new **M12**. The mobile UI is the khelsutra track; the engine must support it (async jobs + status + Drive round-trip / signed URL). Net-new scope (OAuth + Drive write); not necessarily v1, but the design must not preclude it (the `StorageBackend` ABC accommodates it). |
42 | D15 | **Auto-detect SUGGESTS, the user CONFIRMS ? execution is NEVER a surprise.** A GPU owner may still choose cloud; **cloud (= spend) is never entered silently**; the active profile is always explicit + surfaced. | owner 2026-06-21 | Amends §4.3 (auto-detect = a proposed default, not an auto-decision). |
43 | D16 | **M6/M7 implementation SHAPE + the cheap §8 picks (owner 2026-06-21):** (a) **Cloud Run JOBS** (not Services-with-GPU) for the detect+stitch unit, **bucket-poll for completion** (reuses `JobWaiting`/`claim_due_job` verbatim, fits async upload?poll?reel + scale-to-zero D7); (b) **stitch ENCODE = libx264** (protects the D5 byte-determinism/parity guarantee + the L4 parity test) ? **NVDEC for DECODE only** (a speed lever that never touches output bytes); **NVENC encode is DEFERRED behind a config knob**, revisit only if M7 cost data justifies relaxing parity (§8 Q8); (c) **input cap = hard-reject `<2GB`** for v1 (a config cap so raising to 4GB later is one line; matches the shipped M2 chunked-upload threshold ? §8 Q9). | owner 2026-06-21 | M6/M7. Resolves §8 Q2 (co-located, = D4), Q8, Q9, Q10 (dir name confirmed). |
44 | D17 | **Real cloud verification BUDGET = $100 / ?10k cap (owner 2026-06-21):** the owner authorized actual GCP spend for due-diligence + production-grade testing/verification of

M5?M7 (real L4 run, the M5 runlog, the M7 DEPLOY_REPORT), **provided the cap is respected** ? confirm exact cmd + hourly cost before each spend, prefer the smallest viable run + cheapest-adequate GPU, and **DELETE every resource after** ([[gcp-vm-always-delete]]). Removes the M6/M7 "spend" gate within the envelope; the calibrated pricebook (M8) + counsel ToS (M9) + OAuth app (M12) stay owner/counsel-side. | owner 2026-06-21 | Unblocks the real M5/M7 runs (not just the code) within the cap. |

3. Foundation ? evolution / ADR lineage ? *trace the idea end-to-end*
This project is the latest step in a long decision chain; each ADR contributed a piece and a **subtlety that carries forward**:

ADR / doc	Contributed	Subtlety carried into this project
ADR-005	Permissive licensing (MIT/Apache-2.0/BSD only)	The engine + any served model + the **container image** (ffmpeg, fonts, weights) stay commercial-clean.
ADR-008	Owned-model topology; **train==serve featurizer single-sourcing**	The "core unaffected across profiles" is the same parity discipline, extended from train/serve to **local/cloud**.
ADR-009	KhelSutra Guru; **local-first delivery**	The **truly-local profile** is the direct continuation ? privacy/offline self-host stays first-class.
ADR-010	DECOUPLED: DO?GCP pivot	The cloud-compute origin (GPU + co-located bucket).
ADR-011	DECOUPLED scope = M0?M2+M3.5; **deferred M3 (serving) + M4 (billing/cost-guard)**; overrode the *"rent nothing / not a service"* posture	**This project realizes M3/M4.** Carries §06's owner-gated gates: **DPA/consent for non-owner uploads, collector `md5` keying, WASB-C3**.
ADR-012	GPU-native; TPU deferred; **Cloud Run+GPU scale-to-zero = serving model**; **NVDEC** named as the decode lever	The serving substrate + the 7-step seed scope this expands.
ADR-013	Drop DO; **GCP sole compute stack**; $200 DO credits ? non-compute only	The provider for cloud-serving; DO Spaces remains only a possible S3 *storage* target.
? ADR-014 (new)	**Compute-decoupled serving:** one pure core, profile-selected compute+storage; stitch-where-metered	This doc. **Refines** ADR-011/012, does **not** supersede them.
PLATFORM_ARCHITECTURE.md	The `ComputeBackend`/`StorageBackend` registries	The abstraction this realizes in code.
VIDEO_LOCALITY_MODEL.md / DESKTOP_APP_PLAN.md	L0 fully-local tier / `LocalDesktop` backend	The truly-local profile = these, productionized.
07-COMPUTE-ABSTRACTION (archived)	`DeviceContext`/`DeviceProfile` (designed)	WASB hardcodes `device='cuda'` with **no CPU fallback** ? a portability gap M4 closes.
DEPLOY_REPORT_GCP_2026-06-20 (archived)	First real L4 run; **cross-GPU sub-pixel parity finding**	Why D5 (parity at window level, not CSV bytes).

4. Design / architecture
See [`architecture.svg`](architecture.svg) for the picture. **The core never changes; the box around it does.**

4.1 The core contract (must stay byte/behaviour-identical) `[verified]`
1. **DETECT (GPU):** `DetectorRunner.run_predict(video_path, output_dir, video_id) ? CSV[Frame,Visibility,X,Y]` (`backend/pipeline/detectors/base.py:164`). Three interchangeable impls ? `WasbRunner` (WSL), `NativeWasbRunner` (Linux GPU), `StubDetectorRunner` (CPU mock) ? emit the **identical CSV schema**. The trajectory CSV is the stable artifact boundary.
2. **WINDOW (CPU pure fn):** `trajectory_to_action_windows(points, fps, frame_width, ?) ? windows` (`tracknet_runner.py:281`). A shared `@staticmethod`, zero GPU/IO/randomness ? same trajectory + config = byte-identical candidate windows. Used by every runner *and* the eval/regression stack verbatim.
3. **STITCH (CPU/ffmpeg):** `VideoCompiler.compile_highlights(raw_video, segments, out_name) ? reel` (`stitcher/compiler.py:303`). Pure FS+subprocess (merge ? ffprobe ? watermark ? per-clip libx264 trim ? concat). Deterministic for a given input + ranges + watermark config.

Non-goal (the trap to avoid): *no* code branch inside detect/window/stitch keyed on profile ? the same discipline the experiment harness already enforces (a disabled cue is byte-identical to CONTROL).

4.2 The deployment profiles

Profile	Compute	Storage	Orchestration	When
truly-local	`compute_target=local_gpu`; `detector_impl=native` (Linux) / `auto` (WSL); stitch **in-process**	`local` (or `gdrive` = mounted Drive)	`python -m backend.main` or `worker infer` CLI; existing async job worker	Self-host / offline / privacy / dev-on-a-GPU-box ? owner (b)
cloud-serving	`compute_target=cloud_run`; `detector_impl=native` (L4, cuda); **detect AND stitch on the same Cloud Run job**	`gcs` (or gdrive mount / assembled upload); per-video workspace; reels via signed URL	Cloud Run control plane enqueues ? Cloud Run GPU job pulls/runs/pushes; `WAIT_AND_RESUME` reschedules the control-plane job; scale-to-zero	Hosted SaaS ? owner (a)
cpu-mock	`compute_target=cpu_mock`; `detector_impl=stub` (synth/replay)	`local`/in-memory fakes	`run_all_tests.py` / pytest	CI, regression, profile-parity, GPU-free dev
byo_worker (deferred)	tenant GPU via outbound pull agent	tenant's own bucket (signed URLs)	long-poll pull; same job table scoped by tenant	Privacy-sensitive enterprise ? **DEFERRED**, enum/seam reserved

4.3 Profile selection (the "startup/setup config") `[design]`
ONE new top-level `deployment` section on `AppConfig` + preset application + env auto-detect in `load_config`:
- `deployment.profile ? {truly-local | cloud-serving | cpu-mock}` (empty = today's manual knobs, **fully backward-compatible**).
- Selecting a profile applies a **coherent preset bundle** of *existing* knobs that today drift independently: **INPUT** (new `input_backend`/`input_root`, parallel to the output-side `StorageConfig`), **COMPUTE** (new `compute_target` enum locking `detector_impl` + `skip_ai_handoff` + workspace strategy), **STORAGE** (`storage.backend`/`workspace_root`/`single_folder_per_video`, all already exist).
- **Precedence:** built-in defaults ? profile preset ? `config.json` ? env (`RALLY_DETECTOR_IMPL`, `WASB_*`, `DEPLOYMENT_PROFILE`, ?) ? worker `--set k=v`. Every knob stays individually overridable.
- **Env auto-detect SUGGESTS, the user CONFIRMS (D15 ? no surprise execution):** the env gives a *proposed* default (`K_SERVICE` ? cloud-serving; `CI=true` ? cpu-mock; else `torch.cuda.is_available()` ? truly-local), but the user **explicitly confirms or overrides** it ? a GPU owner may still pick cloud, and **cloud (= spend) is NEVER entered silently**. The active profile is always surfaced (logged/shown). **Per-profile startup checks fail/warn fast** (cloud-serving without GCS creds/GPU ? loud error before any job).
- The worker (`cmd_infer`/`cmd_train`) is **already profile-agnostic** (reads `config.storage`, accepts `--config`/`--set`) ? no worker rewrite.

4.4 Compute placement ? and *why stitch runs on Cloud Run* `[design]`
DETECT on GPU everywhere; WINDOW CPU-pure everywhere; STITCH is real CPU work. In **cloud-serving, stitch is co-located in the Cloud Run job** so `gpu_active_seconds` (detect) + `wall_seconds` (detect+stitch) + `bytes_out` (reel egress) land on **one metered invocation** ? the per-job cost ledger. Running stitch on a laptop would leave that cost **invisible/unattributed** ? exactly what the owner wants to avoid. (Stitch stays synchronous in the GPU job for the common one-reel case; a queued CPU-only stitch fallback only if compiles get bursty ? the job table + `claim_due_job` already supports both.)

4.5 Reuse map ? most of the seams already exist `[verified]`
Exists: the 3-stage core; `_resolve_runner` (the single compute seam); `StubDetectorRunner` (`replay_csv` = byte-exact $0 anchor); `StorageBackend` ABC + 4 backends + `StorageRef.parse_uri`; the worker CLI (fetch/put, `--config`/`--set`); the async job control plane (`jobs` table, `claim_due_job` CAS); `ExecutionPolicy WAIT_AND_RESUME`/`JobWaiting` (carries to Cloud Run dispatch verbatim); the chunked-upload router (`<2GB`); `skip_ai_handoff` (LocalNoAI); storage fakes; golden fixtures + experiment harness. **Partial:** per-video workspace helper (default-OFF); `DeviceContext` (designed); edge-auth (v4 `user_id` columns exist). **Net-new:** the `deployment` section; `input_backend`; configurable output base (kill hardcoded `output/`); Cloud Run dispatch shim + container image; cost ledger + pricebook; edge-auth header extraction; startup env detection.

4.6 The data-flywheel harness (D12 / M11) `[design]`
Gemini-on (D11) ships good reels now; this harness is **how we earn the right to flip Gemini off**. On a **consented, adult** cloud-serving job (the D10 trade), the pipeline additionally:

98 1. **Captures** the source upload + the Gemini reel + the rally output (intervals/report) into the per-video workspace (`workspaces/<video_id>/`), instead of the no-consent auto-delete path.
99 2. **Reliably stores** them durably in the bucket (the consented-retention bucket; raw video kept only for the training pool, not the no-consent path).
100 3. **Runs shadow evals** ? the owned model scores the same video alongside Gemini (`backend/eval/experiment.compare` + the dual-eval-set, [[eval-dual-set-regression]]) ? tracks the owned-vs-Gemini gap per job (the live read on "is the floor reachable yet?").
101 4. **Promotes to golden** ? captured videos that pass review become **golden TRAINING rows** (human-reviewed labels via the annotation flow ? `data_pipeline/`), growing the corpus ? the owned model climbs to the D11 floor ? flip Gemini off via a Launch Report.
102 This is the consent?capture?eval?goldenize **flywheel** (the cloud realization of the PERSONALIZATION contribution model): every consented free reel makes the owned model better. Reuses `data_pipeline/`, `backend/eval/`, and the M9 consent gate; **adults-only**, raw-retention gated by the D10 consent.
103
104 ## 6. Honest limits ? what this does NOT do / prove
105 - A green CI suite proves **plumbing + determinism (window+stitch) + profile-parity** for $0 ? it does **NOT** prove field quality, nor that real-GPU detection is good (that's owner-side, real-video).
106 - **Cross-GPU detection is NOT byte-identical** ? parity is a window-level claim, not a CSV-byte claim.
107 - The cloud `0.1.0-l4` weights are **n=2 plumbing-grade** ? cloud-serving ships with **Gemini handoff ON** until the owned model clears a floor (open Q).
108 - This does not resolve **WASB-C3** (sharing a trained model) or the **DPA/consent** gate ? both remain owner/counsel-gated before public, non-owner serving.
109
110 ## 7. Deliverables & progress tracker ? **source of truth**
111 Legend: ? Todo · ? In progress · ? Done · ? Gated. **One small PR per row**, off `origin/master`, no stacking. Default-OFF where it touches core callers.
112
113 | ID | Deliverable | Depends | Gated? | Status | PR |
114 |----|-------------|---------|--------|--------|----|
115 | M0 | This design doc + project dir + `runlogs/` + ADR-014 | ? | owner sign-off | ? | **? APPROVED 2026-06-21** |
116 | M1 | `deployment.profile` + `compute_target` config + preset bundles + precedence/auto-detect; unit tests (no behaviour change, profile='' default) | M0 | safe | ? | [#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279) |
117 | M2 | `input_backend`/`input_root`; wire main CLI/API input through `StorageRef.parse_uri` (local=identity); storage-fake tests | M1 | safe | ? | [#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280) |
118 | M3 | Configurable output/scratch base ? kill hardcoded `output/` (`trajectory_hybrid:237,313`, `yolo_hybrid:407`, `gemini`); **DEFAULT stays `output/`**; golden + stub-E2E byte-identical | M1 | ? default-OFF + Launch Report on flip | ? | [#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282) |
119 | M4 | `DeviceContext` + WASB **CPU-fallback**; unified healthcheck/device wiring | M1 | safe (cuda default unchanged) | ? | [#283](https://github.com/Khelsutra/badminton-highlight-indexer/pull/283) |
120 | M5 | **truly-local profile** end-to-end (preset + startup checks); first committed **runlog** (truly-local sample run) | M2,M3,M4 | owner real-GPU | ? | [#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286) (preset+startup-checks+L4 parity ?; runlog ? owner GPU) |
121 | M6 | Cloud Run GPU **dispatch shim** (control plane ? job via storage ? re-claim) + **containerized worker image** (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests | M5 | ? owner (GCP spend) | ? | [#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287) (shim+image+mocked ?; real build/run = M7 ? owner) |
122 | M7 | **cloud-serving profile** e2e on L4: upload`<2GB` + gdrive/bucket ? detect+stitch on Cloud Run ? reel via signed URL; **DEPLOY_REPORT** (posterity, sample runs + contact sheet) | M6 | ? owner (spend) | ? | ? |
123 | M8 | Per-job **cost ledger** (v6 migration: `owner_id, gpu_active_seconds, wall_seconds, bytes_out, cost_usd, cost_breakdown_json`) + **pricebook** + aggregation + `/api/.../cost` | M7 | ? owner (billing) | ? | [#288](https://github.com/Khelsutra/badminton-highlight-indexer/pull/288) (ledger+pricebook+/cost ? default-OFF; real prices + usage-emission ? owner/follow-up) |
124 | M9 | **Edge auth** (Cf-Access extraction) + per-tenant scoping on `user_id`;

DPA/consent gate wiring | M7 | ? owner (privacy/counsel) | ? | [#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290) (auth: Cf-Access JWT verify + owner_id tag, default-OFF ?; **+ exposure-safety boot posture** in `startup.py` ? edge-auth-on requires the `allowed_video_root` jail + CF team/AUD or the server refuses to start, see changelog; consent cols + ToS/DPA ? owner/counsel) |
M10	`byo_worker` / zero-infra-BYO ? **design-only, DEFERRED**	M8,M9	? explicit owner approval	?	?
M11	**Data-flywheel harness (D12):** on a consented adult job, **capture** the upload + Gemini reel/rally output ? **reliably store** under the per-video workspace ? **shadow-eval** owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? **promote** good ones to the golden TRAINING set (human-reviewed labels via the annotation flow). Closes the loop so the owned model climbs to the D11 floor. Reuses `data_pipeline/` + `backend/eval/` + the consent gate (M9).	M7,M9	? owner (consent/privacy)	?	**skeleton landed** (see changelog): `backend/flywheel.py` + `FlywheelConfig` (default-OFF) + gated `_maybe_run_flywheel` hook, **consent faked-deny**; live activation ? owner/counsel (consent schema) + owned-model/golden data
M12	**`gdrive-api` (OAuth) StorageBackend (D14 Northstar):** cloud reads the source from + writes the stitched reel **back to the user's Google Drive next to the source** (per-user OAuth consent) ? the mobile/cloud-native Drive round-trip. Distinct from the local mount backend; slots into the `StorageBackend` ABC.	M7,M9	? owner (OAuth/consent)	?	**backend landed** (see changelog): `backend/storage/gdrive_api.py` `GDriveApiStorage` + `gdrive-api://` scheme + ADC build; hermetic fake-Drive tests. **Live per-user OAuth app ? owner**

Testing strategy is a first-class deliverable ? [`TESTING_STRATEGY.md`](TESTING_STRATEGY.md). **Run logs / sample runs for posterity** ? [`runlogs/`](runlogs/).

8. Open questions ? owner *(blocking-gates vs nice-to-knows)*
? The 5 gating questions are LOCKED (owner, 2026-06-21) ? see §2 D7?D12:
1. ? **Cold-start ? D7** (scale-to-zero, accept cold-start).
3. ? **Pricebook ? D8** (versioned DB table; USD internal / INR display).
4. ? **Edge auth ? D9** (Cloudflare Access; verify Cf-Access JWT, lock ingress to proxy).
5. ? **DPA/consent ? D10** (free-tier data-for-reels trade; adults-only; derived-data; counsel-gated to M9).
6. ? **Quality floor ? D11** + the **data-flywheel harness D12 / M11** (Gemini-ON until the owned model clears the floor; capture?store?eval?goldenize meanwhile).

? The remaining smaller picks are LOCKED (owner, 2026-06-21) ? see §2 D16:
2. ? **Stitch placement ? co-located** in the detect job (= D4; one metered unit; the job table supports a future CPU-split non-breaking, revisit only on bursty-compile metrics).
7. ? **RESOLVED ? D14:** cloud needs a **`gdrive-api` (OAuth) StorageBackend** (read the source from + write the reel back to the user's Drive) ? the Northstar's Drive round-trip ? distinct from the local **mount** backend (which stays for truly-local). New milestone **M12**.
8. ? **Codec ? D16:** **libx264 ENCODE** (protect parity) + **NVDEC DECODE-only** (speed, no output-byte change); NVENC-encode deferred behind a config knob.
9. ? **Input cap ? D16:** **hard-reject `<2GB`** for v1, as a config cap (raise to 4GB later in one line).
10. ? **Directory name ? `COMPUTE_DECOUPLED_SERVING/` CONFIRMED final** (already the name everywhere + in ADR-014); `CLOUD_RUN_GPU_SERVING/` dropped.

9. Definition of done
- [ ] M1?M4 land (config + input + output-base + device layer), each green on golden cross-OS + experiment no-regression.
- [ ] **Profile-parity test green** ? truly-local vs cloud-mock produce byte-identical windows + segment ranges.
- [ ] **truly-local** (M5) and **cloud-serving** (M7) each have a committed **runlog / DEPLOY_REPORT** with a real sample run.
- [ ] Cost ledger (M8) makes GPU + **stitch** + egress cost visible per job; edge auth (M9) gates multi-tenant.
- [ ] **Data-flywheel harness (M11)** captures?stores?shadow-evals?goldenizes consented adult jobs; the owned model is on a measured path to the D11 floor.
- [ ] ? flip Status DONE, archive to `docs/archives/past_projects/`.

154 ## 10. References
155 **ADR lineage:** ADR-005/008/009/010/011/012/013 ? **ADR-014** (`docs/archives/decisions/DECISIONS.md`). **Strategy:** `PLATFORM_ARCHITECTURE.md`, `HOSTING_PLAN.md`, `VIDEO_LOCALITY_MODEL.md`, `DESKTOP_APP_PLAN.md`, `LOCAL_APPLIANCE_ARCHITECTURE.md` (khelsutra). **Archived predecessor:** `docs/archives/past_projects/DECOUPLED_COMPUTE/` (esp. `05-COST-ATTRIBUTION.md`, `07-COMPUTE-ABSTRACTION.md`, `06-RISKS-AND-OPEN-QUESTIONS.md`, `DEPLOY_REPORT_GCP_2026-06-20.md`). **Code anchors `[verified]`:** `backend/pipeline/detectors/base.py:164`, `tracknet_runner.py:281`, `stitcher/compiler.py:303`, `segmenters/trajectory_hybrid.py:65,237`, `backend/storage/*`, `backend/worker/__main__.py:88`, `backend/api/{uploads,job_worker}.py`, `backend/infrastructure/database.py`. **Design provenance:** workflow `wf_56274ff0` (4 code-mappers ? synthesis; 58 findings).
156
157 ---
158 ### Changelog
159 - 2026-06-21 ? created from the design workflow output; M0 in progress.
160 - 2026-06-21 ? **M0 APPROVED**; **M1 landed** ([#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279)): `deployment.profile` + `compute_target` + preset bundles + precedence/auto-detect-as-suggestion (D15) on `AppConfig`. Pure/additive, `profile=''` default == today's behaviour (zero core change); presets touch only existing knobs (no M2 `input_backend` / M3 output-base over-reach). Suite 1068 green, cov 84.11%.
161 - 2026-06-21 ? **M2 landed** ([#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280)): `input_backend`/`input_root` + URI-aware main API/CLI input via `resolve_input_uri`/`acquire_local_input` (local = identity passthrough, byte-for-byte unchanged; bucket/Drive fetched to scratch). Unsupported scheme ? clean 400; `local://` accepted on the resolved key; hosted-mode rejects non-local URIs; source URI kept for DB/report provenance. Suite 1105 green, cov 84.84%.
162 - 2026-06-21 ? **M3 landed** ([#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282)): configurable output/scratch base via `backend/utils/paths.output_base(config)` ? the 3 segmenters' chunk/handoff/temp dirs root under `output.output_dir` (default `"output"`, byte-identical; default-OFF ? no Launch Report). Fixes the latent bug where scratch ignored `--output-dir`. Suite 1114 green, cov ~84.9%.
163 - 2026-06-21 ? **M4 landed** ([#283](https://github.com/Khelsutra/badminton-highlight-indexer/pull/283)): `DeviceContext` + native WASB **CPU-fallback** via a new `device="auto"` (cuda-if-available-else-cpu); explicit `"cuda"` stays strict (no silent slow-CPU downgrade), default unchanged. `"auto"` resolves before argv (never reaches `wasb_infer.py`); CUDA probe cached. Device-selection tested offline; real CPU-inference run owner/GPU-verified. Suite 1126 green, cov 84.96%. **? M1?M4 (the config/input/output/device layer) COMPLETE.**
164 - 2026-06-21 ? **M5+ batch authorized + design picks locked** (owner): the remaining §8 picks resolved ? **D16** (M6 = Cloud Run **Jobs** + bucket-poll; **libx264 encode + NVDEC decode-only**; **hard-reject `<2GB`** input cap; dir name confirmed) and **D17** (real cloud verification budget = **$100/?10k cap**, delete-after). Owner gave a blanket go to write+merge the **default-OFF code** for every remaining gated milestone (M5,M6,M8,M9-auth,M9-consent-cols,M11-plumbing,M12), one isolated PR each, reserving the gate for the real run/spend/calibration/counsel. Grounded by gate-map workflow `wf_4849400f`.
165 - 2026-06-21 ? **REAL M7 cloud GPU run ?** (owner-authorized, within the D17 cap): provisioned an **L4** (`g2-standard-4`, asia-southeast1-a; Mumbai stocked-out) ? native WASB on the L4 ? **22 rallies** on the smoke clip ? `gs://?/outputs/05e0e577?/{intervals.csv,report.json}`; VM DELETED ($0, ~$0.21). ? the no-Gemini cloud run needs `--set indexing.skip_ai_handoff=true` (the fully-local path emits WASB candidate windows directly ? the long-flagged "0 rallies" cold-start fix). The reel **stitch** validated locally (`VideoCompiler` ? 22-segment highlights, 85 MB). DEPLOY_REPORT runlog = the cloud-GPU dry-run guide (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`).
166 - 2026-06-21 ? **M9-auth code landed** ([#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290)): `backend/api/auth.py` ? Cloudflare Access **Cf-Access JWT verify** (RS256: sig+exp+iss+aud against the team JWKS; lazy `PyJWT[crypto]`; injectable JWKS ? fully-offline tests) + `resolve_tenant` ? tags `create_job(owner_id=?)`; `SecurityConfig.{edge_auth_enabled=False,cf_team_domain,cf_access_aud}` + CORS via `RALLY_CORS_ALLOW_ORIGINS`. **Default-OFF.** Security review (`wf_9b4c71d0`) found NO bypass; locked alg=none + HS256-confusion rejection. **? M9 ? ? the consent cols + ToS/DPA are owner/counsel; the consent migration is now v7 (M8 took v6).**
167 - 2026-06-21 ? **M8 code landed** ([#288](https://github.com/Khelsutra/badminton-highlight-indexer/pull/288)): per-job **cost ledger** ? `backend/billing.py` (pure `usage × pricebook` math, USD internal / INR display) + a **v6 migration** (NULLABLE cost columns on

`jobs` + a versioned `pricebook` table seeded `placeholder-v0` at $0; additive-NULL, idempotent) + `record_job_cost`/`get_job_cost`/`get_video_cost` + `GET /api/jobs/{id}/cost` + `/api/videos/{id}/cost` + `BillingConfig` (**default-OFF**) + a gated `_maybe_record_job_cost` hook. Adversarial review (`wf_498b24af`) folded a **major silent-GPU-under-bill** (an unpriced `gpu_type` is now surfaced, not invisible $0) + the **honest no-op** caveat (usage emission is a follow-up ? cost 0 until wired) + breakdown reconciliation. Suite green, ruff+mypy clean. **? M8 ? ? owner inserts real GCP prices (a new pricebook version) + the worker usage-emission is a follow-up; placeholder ships $0 = no user-facing change. v6 is the shared bump (M9-consent cols extend it).**

168   - 2026-06-21 ? **M6 code landed** ([#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287)): `backend/compute/` ComputeBackend registry ? `CloudRunCompute` dispatches a Cloud Run **Job** (injectable client) then **bucket-polls** a done-marker via `execution_policy.poll_until` + the storage layer (D16); `get_compute_backend` returns a backend ONLY for `compute_target=cloud_run` (else `None` ? the worker's in-process path, **default-OFF**). The dispatched container runs **inline** (forced `local_gpu`+native, never re-dispatch); the worker writes a `--done-uri` completion marker on success. `deploy/cloudrun/{Dockerfile,README}` = the L4 image (CUDA+ffmpeg+fonts+`wasb` conda env; weights staged per WASB-C3) + the M7 runbook. Adversarial review (`wf_87b22fa4`) folded a **blocker** (cloud `fetch` needs a dst ? reads silently returned `{}` on gcs/s3) + a poll-wait test gap + `PolicyTimeout`?clean rc 4. Suite green, ruff+mypy clean. **? M6 ? ? the real build/push/L4 run + DEPLOY_REPORT is M7 (owner, D17 budget).**

169   - 2026-06-21 ? **M5 code landed** ([#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286)): `backend/config/startup.py` ? per-profile fail/warn-fast startup checks (cloud-serving + non-cloud storage = the one fatal coherence error; GPU/creds mismatches = warnings, the runner healthcheck stays the authority), wired into the server (`_enforce_startup_or_exit` ? exit 1) + the worker (`cmd_infer`/`cmd_train` ? rc 2), default-OFF (probe gated behind an active profile ? the worker stays torch-free at `profile=""`). **L4 profile-parity test** (`tests/test_profile_parity.py`): same stub trajectory ? byte-identical windows + segment ranges across truly-local / cpu-mock / no-profile + worker-vs-in-process. Adversarial 4-lens review folded (`wf_34671ac1`). Suite green, ruff+mypy clean. **? M5 ? ? the real-GPU truly-local runlog (L5) is owner-side** (template pre-staged in `runlogs/`).

170   - 2026-06-21 ? **M9 hardening: exposure-safety boot posture** (alpha-shareable "tunnel-to-a-box" enabler): extended `backend/config/startup.py` with a **server-only exposure posture** (new `include_exposure=True` from `_enforce_startup_or_exit`; the worker stays unaffected ? it reads `--video-uri`, not untrusted tenant paths). Gated on `security.edge_auth_enabled` (**default-OFF ? strict no-op**, byte-identical to today): when the owner flips edge auth ON to expose the engine behind the CF-Access gate, three checks fire at **boot** ? `EXPOSED_NO_VIDEO_ROOT` (**fatal**: `allowed_video_root` jail unset ? a tenant could read an arbitrary local file), `EXPOSED_EDGE_AUTH_INCOMPLETE` (**fatal**: CF team-domain/AUD missing ? lifts the current first-request 401 to boot), `EXPOSED_AUTH_EXTRA_MISSING` (**warn**: `PyJWT[crypto]` not importable ? would 500 per request). So a half-configured exposure fails fast instead of leaking/500-ing. +15 unit tests (default-OFF no-op, worker-ignores, both-fatals-compose-with-profile-checks, server-exit, WARN-only-no-exit). Adversarial 3-lens review (`wf_35b735a9`) confirmed default-OFF byte-identical + no bypass; folded a **whitespace-drift** (aligned `auth.resolve_tenant` to `.strip()` team/AUD so "configured" means the same at boot + at request time) + named the config fields in the edge-auth-incomplete message. Pairs with the appliance **T3** network posture (raw `:8000` refused ? owner-run). Suite green, ruff+mypy clean. **? backstops M9 ? the actual exposure flip + CF team/AUD/ingress + key rotation stay owner-side. (Deferred to the PR-2 runbook: document that `upload_dir` must sit under `allowed_video_root`, and that you MUST set `edge_auth_enabled=True` when exposing ? these checks only fire then.)**

171   - 2026-06-22 ? **M11 skeleton landed** (data-flywheel harness, D12): `backend/flywheel.py` (`capture_job` ? per-video `VideoWorkspace` via `backend.storage.put`; `shadow_eval` ? `experiment.compare`/`compare_unlabeled`; `propose_promotion` ? `promotion.promote_if_served_safe` with Gemini=control/owned=proposal; `run_for_job` orchestrator) + `FlywheelConfig` (**default-OFF**) + a gated `_maybe_run_flywheel` job-completion hook mirroring M8's cost hook. **TWO independent guards keep it INERT in production:** `flywheel.enabled=False` (default) AND `consent_attested` is a **FAKED hard-deny** (returns False for every job) ? so even enabled it captures **nothing** until the real consent schema lands. This is the PLUMBING only; it reuses the real workspace/eval/promotion/storage code so it works the day consent + the owned model + golden labels are wired. **NO DB migration** (the skeleton persists no consent). +13 tests (default-OFF no-op, enabled-but-consent-denied no-op, consented-capture writes the manifest, delegation of shadow-eval/promote, the hook is best-effort). Suite green, ruff+mypy clean. **? M11 ? ? LIVE activation is owner/counsel (the consent ToS/DPA + schema ? the consent columns) + owner/GPU (the owned model

+ golden corpus for shadow-eval/promotion). ? the future M9-consent migration is now `v9` (v7 = `locale`, v8 = the compute-picker `compute` column ? the schema ladder drifts as siblings land; check `PRAGMA user_version` before authoring).**
172   - 2026-06-22 ? **M12 backend landed** (`gdrive-api://` OAuth/API StorageBackend, D14 Northstar): `backend/storage/gdrive_api.py` `GDriveApiStorage(StorageBackend)` ? the headless **Drive-API** round-trip (read source + write reel back), distinct from the mount-only `gdrive://` (left untouched). Path-keyed `gdrive-api://<path>` resolves each component ? a Drive file id via `files().list`; implements `fetch_to_local` (`get_media().execute()` bytes), `put` (folder-creating `files().create` + a `_media_body` seam), `exists`/`stat`/`list`/`delete`. Wired the scheme into `ref.py` (`_KNOWN` + the path-keyed branch), `__init__._backend_class`, and the `StorageConfig.backend`/`input_backend` Literals + a `gdrive_api` settings dict (read via a **hyphen?underscore** lookup in `get_storage`). **All Google libs lazy-imported** (`import backend.storage` never pulls them); a `service` is **injectable** ? fully-offline tests via an in-memory fake Drive (round-trip + scheme + registry + the underscore-settings lookup); `_build_service` (ADC over the Drive scope) raises a clear `storage-gdrive-api`-extra error when libs/creds are absent. **NO DB migration.** Adversarial 2-lens review (`wf_ae8c9b0f`). Suite green (1466), ruff+mypy clean. **? M12 ? ? the LIVE per-user interactive OAuth app (consumer "connect your Drive": client id/secret + refresh tokens) is owner-gated; the backend ships the service-account/ADC build + the fake. Reads no secrets from config (creds via ADC only).**
173

9. assistant (2026-07-03T16:40:18.659Z)

Now let me search for specific reuse and duplication patterns in the new document against these existing docs:

10. user (2026-07-03T16:40:19.850Z)

docs\COMMERCIALIZATION.md:6:**Purpose.** Single source of truth for the **commercial** dimension of the project:
docs\CODE_MAP.md:128:- `@backend/config/models.py` ? Pydantic v2 models; **single source of truth for defaults**.
docs\CODE_MAP.md:177:[Omitted long matching line]
docs\CODE_MAP.md:289:- **Segmenter default:** single source of truth = model `IndexingConfig.default_segmenter='yolo_hybrid'`; `config.json` may override at runtime (ships `gemini`). run_*.py read the model value (no literals); `main.py` keeps the defensive `'motion'` getattr fallback noted above.
docs\CODE_MAP.md:315:| Change a config default / add a config field | `@backend/config/models.py` (single source of truth) -> regenerate via `scripts/doctor.py`/`save_default_config`; ? config.json may override (default_segmenter) |
docs\ALPHA_LAUNCH_READINESS.md:5:> **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed in `docs/README.md` and `docs/DOC_STATUS.md`.
docs\ALPHA_LAUNCH_READINESS.md:91:## 7. Deliverables & progress tracker
docs\DECISION_SNAPSHOT_REGRESSION.md:3:Follows docs/PROJECT_DOC_TEMPLATE.md. §7 progress tracker is the source of truth.
docs\DECISION_SNAPSHOT_REGRESSION.md:10:> **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`; pointer in memory `current-state` once work starts.
docs\DECISION_SNAPSHOT_REGRESSION.md:60:## 7. Deliverables & progress tracker ? **source of truth**
docs\DESKTOP_APP_PLAN.md:5:> **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once the owner greenlights P0).
docs\DESKTOP_APP_PLAN.md:124:## 7. Deliverables & progress tracker   ? **source of truth**
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:8:This document captures an end-to-end investigation and planning exercise. It is intended to be **updated in place** as work progresses (findings, test additions, verification runs, conclusions). It serves as the single source of truth for the hardening effort so future PRs have a clear baseline and checklist.
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:50:- **Impact:** Every validator/segmenter/detector is coupled to the legacy dict shape. The "single source of truth" model is aspirational today.
docs\DOC_STATUS.md:11:- **Marking something complete?** Update the doc's own **§7 progress tracker** (the source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2 archival checklist.

```
docs\GOLDEN_REGRESSION_FIXTURES.md:5:> **Tracking anchors:** §7 progress tracker is the source
of truth; indexed in `docs/README.md`; pointer in memory `current-state` (and
`docs/NEXT_STEPS.md` once work starts).
docs\GOLDEN_REGRESSION_FIXTURES.md:145:## 7. Deliverables & progress tracker  ? **source of
truth**
docs\COMPUTE_DECOUPLED_SERVING\TESTING_STRATEGY.md:3:> **Companion to** [`README.md`](README.md)
(§7 tracker is the source of truth). **Status:** DRAFT (2026-06-21).
docs\I18N_PLAN.md:69:  en/  common.json   # source of truth ? devs author keys here
docs\I18N_PLAN.md:80:- `en` is the source of truth; **every other locale is keyed identically**
(enforced by the
docs\data_pipeline\GOLDEN_VIDEOS.md:3:**Purpose.** The single source of truth for every
**golden** (human rally-boundary-labeled) video ? what
docs\NIGHTLY_ENGINE_REGRESSION.md:5:> **Tracking anchors:** §7 is the source of truth; pointer
in memory `current-state`.
docs\NIGHTLY_ENGINE_REGRESSION.md:69:## 7. Deliverables & progress tracker  ? source of truth
docs\PACKAGING_PLAN.md:5:> **Tracking anchors:** §7 progress tracker is the source of truth;
indexed in `docs/README.md` + `docs/NEXT_STEPS.md`; memory `[[packaging-app-plan]]` + `[[next-
session-packaging-app]]`.
docs\PACKAGING_PLAN.md:95:## 7. Deliverables & progress tracker ? source of truth
docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md:14:Use this as the `prompt`
argument to `scheduler_create` (or paste as the first message in a brand-new session). It is
deliberately short at the top so the MD is the source of truth.
docs\PER_VIDEO_WORKSPACE.md:5:> **Tracking anchors:** §7 progress tracker is the source of
truth; pointer in memory `current-state`.
docs\PER_VIDEO_WORKSPACE.md:77:## 7. Deliverables & progress tracker  ? **source of truth**
docs\PROJECT_DOC_TEMPLATE.md:10:- §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per
independently-shippable PR.
docs\PROJECT_DOC_TEMPLATE.md:23:> **Tracking anchors:** §7 progress tracker is the source of
truth; indexed in `docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md`
once work starts).
docs\PROJECT_DOC_TEMPLATE.md:54:## 7. Deliverables & progress tracker   ? **source of truth**
docs\COMPUTE_DECOUPLED_SERVING\README.md:5:> **Tracking anchors:** §7 progress tracker is the
source of truth; pointer in memory `current-state` + `docs/NEXT_STEPS.md` once work starts.
docs\COMPUTE_DECOUPLED_SERVING\README.md:35:| D8 | **Pricebook = a versioned DB table**; cost
computed in **USD** (matches GCP billing = one source of truth), **displayed in INR** at a
configurable FX rate (Bangalore market); re-versioned when GCP prices change. | owner 2026-06-21
| M8. |
docs\COMPUTE_DECOUPLED_SERVING\README.md:110:## 7. Deliverables & progress tracker  ? **source
of truth**
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:4:>
**Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived`. **Tracking anchor:** the **[Remediation
tracker (live)](#remediation-tracker-live)** section below is the source of truth for
completed/open items; indexed in `docs/README.md`; lifecycle row in `docs/DOC_STATUS.md`; live
checkpoint in memory `current-state`.
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:617:>
Folded in from the former standalone `KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN.md` (archived
2026-07-02 ? `docs/archives/past_projects/KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-
FOLDED-2026-07-02.md`). This is the **source of truth for completed/open items**; the findings
above are the point-in-time snapshot. One small PR per tracker row, no stacking.
docs\archives\past_projects\AUDIT_REPORT_khelsutra-
guru_2026-07-02-COMPLETED-2026-07-02.md:734:### Deliverables & progress tracker  ? **source of
truth**
docs\PROMINENT_COURT_DETECTION.md:5:> **Tracking anchors:** §8 progress tracker is the source of
truth; pointer in memory `current-state`.
docs\PROMINENT_COURT_DETECTION.md:97:## 8. Deliverables & progress tracker ? source of truth
docs\archives\past_projects\KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md:3:> **?
ARCHIVED / FOLDED 2026-07-02** ? this standalone tracker was folded into the umbrella
`docs/AUDIT_REPORT_khelsutra-guru_2026-07-02.md` as its *Remediation tracker (live)* section,
which is now the source of truth. Kept here for lineage. Relative links below are historical-by-
design.
docs\archives\past_projects\KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md:7:>
**Tracking anchors:** Section 7 progress tracker is the source of truth; indexed in
`docs/README.md`; lifecycle row in `docs/DOC_STATUS.md`; live checkpoint in memory `current-
state`.
docs\archives\past_projects\KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md:157:##
```

7. Deliverables & progress tracker
docs\archives\past_projects\CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md:12:> **Tracking anchors:** §6 progress tracker is the source of truth; indexed in `docs/README.md`.
docs\archives\past_projects\CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md:132:- `backend/config/models.py` ? `AppConfig`, `IndexingConfig`, `ValidationConfig` (single source of truth for defaults + types)
docs\archives\past_projects\CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md:176:## 6. Deliverables & progress tracker ? **source of truth**
docs\RALLY_DETECTION_FIXES_PLAN.md:9:> **Tracking anchors:** §7 progress tracker is the source of truth; tracked in `docs/DOC_STATUS.md` §3a.
docs\RALLY_DETECTION_FIXES_PLAN.md:69:## 7. Deliverables & progress tracker   ? **source of truth**
docs\RALLY_DETECTION_GAP_CLOSURE.md:5:> **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory `current-state`; add to `docs/README.md` + `docs/NEXT_STEPS.md` when work starts.
docs\RALLY_DETECTION_GAP_CLOSURE.md:69:## 7. Deliverables & progress tracker ? **source of truth**
docs\README.md:2:[Omitted long matching line]
docs\README.md:3:[Omitted long matching line]
docs\README.md:11:[Omitted long matching line]
docs\README.md:18:[Omitted long matching line]
docs\README.md:19:[Omitted long matching line]
docs\README.md:21:[Omitted long matching line]
docs\README.md:37:- PROJECT_DOC_TEMPLATE.md ? Reusable skeleton for a **tracked project doc**: status header · decisions-locked · progress tracker (?/?/?) · definition-of-done · archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7 and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.
docs\README.md:45:[Omitted long matching line]
docs\archives\README.md:7:- Everything here is no longer the primary source of truth. It is kept for context, decision history, and regression detection.
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:103:- No single source of truth for the config shape.
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:129:The models are the single source of truth for defaults. `setup.py` will call the new save function. `main.py` will use the typed loader early.
docs\archives\past_projects\DECOUPLED_COMPUTE\07-COMPUTE-ABSTRACTION.md:102: header. One source of truth for "what GPU am I on."

11. user (2026-07-03T16:40:20.421Z)

docs\ANALYZERS_AND_RECONCILERS.md:257:a precision target). So an `in_warmup` feature is directly A/B-able on the 6 golden videos for **precision**
docs\ANALYZERS_AND_RECONCILERS.md:302:compare(golden_videos, CONTROL, fusion_plus)   # LOVO-honest B ? A on the 6 golden videos
docs\ANALYZERS_AND_RECONCILERS.md:387:`RECONCILED` variant lets `compare(CONTROL, RECONCILED)` report the LOVO lift on the 6 golden videos
docs\ANALYZERS_AND_RECONCILERS.md:544: LOVO on the 6 golden videos; confirm the owner's **trailing-dead-time misfire** truncates, the
docs\ALPHA_LAUNCH_READINESS.md:13:The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has 15 badminton videos; `[analysis]` the local manifest needed path rebasing before manifest-driven commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion schema; `[analysis]` the n=15 quality baseline is stale and must be intentionally refreshed before it can gate anything.
docs\ALPHA_LAUNCH_READINESS.md:22:- **Can the 15-video golden corpus now support honest quality gates?**
docs\ALPHA_LAUNCH_READINESS.md:34:| D4 | **Do not promote served Gate A by default from the current n=15 evidence.** | `backend.eval.serve_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-windowing safety; keep it as an opt-in/analysis lever. |
docs\ALPHA_LAUNCH_READINESS.md:51:- `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says `output/human_lovo_manifest.json` is the ingested golden corpus at `n = 15`.
docs\ALPHA_LAUNCH_READINESS.md:99:| A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer` | A0 | No | TODO | - |

docs\ALPHA_LAUNCH_READINESS.md:101:| A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2 | Reviewer sign-off | TODO | - |
docs\ALPHA_LAUNCH_READINESS.md:102:| A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6` floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | TODO | - |
docs\ALPHA_LAUNCH_READINESS.md:103:| A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |
docs\ALPHA_LAUNCH_READINESS.md:106:| A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner approval for M0.4 scope | TODO | - |
docs\ALPHA_LAUNCH_READINESS.md:117:4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor refresh.
docs\ALPHA_LAUNCH_READINESS.md:127:5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every clip, or should the eval loaders support mixed schemas with explicit capability tags?
docs\ALPHA_LAUNCH_READINESS.md:133:- [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely on stale local absolute paths.
docs\ALPHA_LAUNCH_READINESS.md:134:- [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
docs\BACKLOG.md:52:- ? **Phase-1 video collection** ? gather the round-1 golden corpus per `docs/GOLDEN_SET_PHASE1_VIDEOS.md`: **static self-recorded** footage (mirror deployment, not broadcast), 3 sports incl. singles & doubles, across the diversity matrix; ~12 clips/sport with **? held out by match**. **Unblocked.**
docs\ABLATION_LAYER.md:44: failure modes the golden corpus may not contain.
docs\CODE_MAP.md:124:[Omitted long matching line]
docs\CODE_MAP.md:336:| Guard rally-quality nightly (no GPU) | `@scripts/nightly_regression.py::main` ? re-score golden corpus (default + milestone configs) vs `eval_baselines/nightly_baseline.json`; exit 1=regress, 4=stale; `--update-baseline` to re-snapshot after an intended change. Scheduled via `scripts/register_nightly_task.ps1` |
docs\DATA_IN_GCS.md:136:| golden manifest | `gs://?/manifests/golden.json` (§7.3); the **local** litmus still reads `output/human_lovo_manifest.json` | 4.5 KB | ? (2026-06-22, §7.3) ? flipping the litmus to read it is Phase 3 | eval corpus definition |
docs\COMPUTE_DECOUPLED_SERVING\TESTING_STRATEGY.md:10:2. **Quality guarantee** ? real WASB detection on a real GPU is *good enough at the rally level*. **Not** byte-deterministic across GPU architectures (L4 vs 2070 differ sub-pixel ? 684/1195 frames, per the 2026-06-20 deploy report), so it is judged at the **window/rally level**, owner-side, on the golden corpus.
docs\COMPUTE_DECOUPLED_SERVING\TESTING_STRATEGY.md:38:- **How:** owner runs the **golden corpus** on the GPU box and on Cloud Run; compare rally counts/windows; **smoke-test the reel** (`ffmpeg -f null -` full decode exit 0 + a 5-frame contact sheet visually confirming real rallies). Recorded in a **DEPLOY_REPORT** under `runlogs/`.
docs\COMPUTE_DECOUPLED_SERVING\TESTING_STRATEGY.md:43:- **Real-GPU is owner-side only** (the golden corpus + labelled videos live with the owner). It validates that `NativeWasbRunner` actually uses the GPU and that cross-GPU detection diffs are **sub-pixel** ? judged at the window level, never CSV bytes.
docs\archives\decisions\DECISIONS.md:374:**The long-term route (Option B ? the intended destination, once a healthy golden corpus exists).** As we
docs\DOC_STATUS.md:33:| `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created 2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-model/commercial rally-detection gates | No | Execute §7 in order: corpus portability + n=15 baselines, hosted alpha serving, ablation/fusion reruns, then launch GO/NO-GO |
docs\COMPUTE_DECOUPLED_SERVING\README.md:171:[Omitted long matching line]
docs\GOLDEN_REGRESSION_FIXTURES.md:94:[Omitted long matching line]
docs\GOLDEN_REGRESSION_FIXTURES.md:159:[Omitted long matching line]
docs\NEXT_STEPS.md:6:[Omitted long matching line]
docs\NEXT_STEPS.md:12:[Omitted long matching line]
docs\NEXT_STEPS.md:15:2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus training feed on it.
docs\NEXT_STEPS.md:59:1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"` ? produces `*_golden_features.json` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)

docs\NEXT_STEPS.md:103:- **Current status (2026-06-14):** P1 (person cue extraction to reusable `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via hit_detector.
docs\MULTI_SIGNAL_FUSION_PLAN.md:160: labels are the honest measuring stick**, and as the Roora golden corpus grows they can also become
docs\archives\quality-iterations\README.md:21:[Omitted long matching line]
docs\archives\quality-iterations\README.md:22:[Omitted long matching line]
docs\PACKAGING_PLAN.md:78:| **HEAVY / local-GPU** (owner + opt-in power user) | Owner w/ golden corpus + CUDA box | LIGHT + micromamba WASB env (py3.8/torch1.11) via `$WASB_PYTHON`; main torch out-of-band via uv; weights SHA-256-pinned. **Gated on C3 + reproducible config**; ONNX supersedes this env | opt-in "Enable local GPU" step; resumable progress (multi-GB on Indian bandwidth); post-install `torch.cuda.is_available()` validate; falls back to LIGHT. |
docs\data_pipeline\VIDEO_RECORDING_GUIDELINES.md:87:4. How does the owned/licensed golden corpus need to grow to *cover* the capture conditions real users will actually use? (Feeds GOLDEN_SET_VENDORING.md §8.)
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:34:1. **Grow the golden corpus (NEXT_STEPS #1 ? highest leverage)**
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:44:   P3: integrate person features (from the P2 FusionSegmenter + `fusion_features`) into the harness `compare` for LOVO litmus on golden videos; report ?F1 + honest stats/CIs/sign test. Eager/windowed compute is in place from P2. If lift: optional served gating (flag default-OFF).
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:91:| 1 | Grow golden corpus | LOW | Data effort + existing curate; consent guard | Manifest + LOVO/sign test improvement |
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:126:*Parallel / owner-led (not primary agent engineering focus):* Grow the golden corpus (your #1 leverage item), PMF field visits (using the existing kit), repo rename. These run alongside M0 without blocking the quick engineering wins.
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:155:scaffolding (async jobs, storage/compute backends) is rock-solid AND the golden corpus is healthy. Paid Gemini
docs\data_pipeline\GOLDEN_VIDEOS.md:5:Use this to **isolate the golden videos into a dedicated location** and to keep the growing golden set
docs\data_pipeline\GOLDEN_VIDEOS.md:28:## Ingested golden corpus ? `output/human_lovo_manifest.json` (n = 15)
docs\data_pipeline\GOLDEN_VIDEOS.md:57:## How to isolate the golden videos
docs\QUALITY_ITERATIONS.md:62:**Measured (n=15 golden, milestone cap6+R4, LOCAL vs golden; reproduce: `python -m backend.eval.rally_seg_eval --stratify`):**
docs\QUALITY_ITERATIONS.md:213:?_start=2/?_end=1.5; see R2_EVAL_METRIC_DESIGN.md), on the **n=11** golden corpus
docs\QUALITY_ITERATIONS.md:241:A scheduled nightly job re-scores the **golden corpus** (`output/human_lovo_manifest.json`, n=11) under HEAD
docs\QUALITY_ITERATIONS.md:254:- **Default-ADDITIVE ? it does NOT change the promotion gate.** The baseline is tied to the LOCAL golden corpus
docs\QUALITY_ITERATIONS.md:339:neutral on the clean golden set** ? the golden videos have few continuously-active >cap windows, so the
docs\QUALITY_ITERATIONS.md:396:follow-on (needs GPU person/audio feature re-extraction on the golden videos ? now feasible with the GPU grant).
docs\QUALITY_ITERATIONS.md:434:plus `backend/eval/rally_seg_eval.py`, the one-command eval: golden manifest + golden_features `gts` ? window
docs\QUALITY_ITERATIONS.md:467: handoff set are **byte-identical to `wasb_hybrid`** on all 6 golden videos (same candidate windows; handoff = all windows =
docs\QUALITY_ITERATIONS.md:535:### Fusion P3 + P4 MEASURED on the n=6 golden corpus ? person = no lift, audio = promising (not promotable) *(2026-06-14)*
docs\QUALITY_ITERATIONS.md:536:The GPU person-feature extraction finished all 6 golden videos (`fusion_golden.py` ? `output/*_golden_features.json`, the
docs\QUALITY_ITERATIONS.md:586:now the standing guard: it pins the honest measured behaviour on the **present golden corpus** and trips if a
docs\QUALITY_ITERATIONS.md:623:### First end-to-end A/B on the golden corpus (n=6) ? **ARCHIVED** *(2026-06-14)*
docs\QUALITY_ITERATIONS.md:697:**Lift still to be measured** offline on the 6 golden videos via `experiment.compare(CONTROL, fusion)`; the
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:190:[Omitted long matching line]
docs\archives\past_projects\AUDIT_REPORT_khelsutra-

guru_2026-07-02-COMPLETED-2026-07-02.md:494:[Omitted long matching line]
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md:60:| **E3** | Add audio features to `distill_local`; re-eval IoU-0.5 LOVO | E1/E2 + ?2 golden videos | does fused beat WASB-only + trajectory-only distilled? |
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md:85: pose + ? ? rallies), fine-tuned on our golden corpus instead of depending on WASB's pretrained weights ? the
docs\R1_MILESTONE_LAUNCH_REPORT.md:9:> **Why this flip is safe to ship:** measured on the **n=13** golden corpus (the grown set, incl. the two
docs\R2_EVAL_METRIC_DESIGN.md:94:study** ? a 2nd labeler re-labels 2?3 already-golden videos; set `?_start`/`?_end` to ~the 90th-percentile
docs\R2_EVAL_METRIC_DESIGN.md:152: ? the natural home, since `rally_seg_eval` loads the golden manifest + trajectories. The dual-set
docs\R2_EVAL_METRIC_DESIGN.md:157: switches to R2 only after a dedicated **ablation experiment** on the golden corpus shows R2 ranks candidates
docs\R2_EVAL_METRIC_DESIGN.md:190: R2 becomes the promotion gate, run a dedicated **ablation experiment** on the golden corpus showing R2
docs\archives\checkpoints\2026-06-eval-baseline\README.md:15:- Any existing P/R/F1 baselines or cross-validation results on the then-current golden corpus.
docs\RALLY_DETECTION_FIXES_PLAN.md:22:> point (n=15 clips / 595 rallies; cached WASB trajectories + labels from `gs://khelsutra-rally-corpus`,
docs\RALLY_DETECTION_GAP_CLOSURE.md:12:[Omitted long matching line]
docs\RALLY_DETECTION_GAP_CLOSURE.md:26:| D3 | **Golden labels = ground truth; Gemini = reference, not truth** | n=15 corpus | Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden. |
docs\RALLY_DETECTION_GAP_CLOSURE.md:48:[Omitted long matching line]
docs\RALLY_DETECTION_GAP_CLOSURE.md:64:- **n is small.** One golden clip scored so far (GX010141); the corpus is n=15 with several weak clips. Per-video conclusions firm up only as the owner labels more (GX010137 / GX020140 / a large-count clip next).
docs\RALLY_DETECTION_GAP_CLOSURE.md:81:| P7 | **Long-rally (>5s) quality lens** (eval `--min-duration`/`--stratify`) + reel knob (`stitching.min_rally_duration`), both default-OFF. **Measured n=15:** single-court over-fire is short-FP-dominated (seg_ratio 1.43?1.04) but **multicourt over-fires 1.58× even on long rallies** (? court-localization, Lever C) and strict tolP doesn't rise (over-merge-contaminated). See [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) "P7". | ? | No (eval+cfg, default-OFF) | ? measured | this PR |
docs\RALLY_DETECTION_GAP_CLOSURE.md:104:- 2026-06-18 ? **P7 long-rally (>5s) quality lens + reel knob (both default-OFF), measured n=15.** Eval `--min-duration`/`--stratify` (`rally_seg_eval.py`) + product `stitching.min_rally_duration`. Finding: single-court over-fire is short-FP-dominated (seg_ratio 1.43?1.04) but **multicourt over-fires 1.58× even on long rallies** (? needs court-localization, Lever C) and strict tolP is gated by over-merge contamination, not just FPs. Sharpened §3 + added §7 row P7.
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md:1:# First end-to-end A/B experiment on the golden corpus (n=6)
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md:99:First real A/B on the golden corpus ? the harness + C1?C4 produced an **honest, promotable finding
docs\RALLY_DETECTION_QUALITY_REPORT.md:20:held-out lift. We evaluate against an owned golden corpus, now **n=11 labelled matches** (grown from
docs\RALLY_DETECTION_QUALITY_REPORT.md:132:### 3.1 The owned golden corpus and the oracle contract
docs\RALLY_DETECTION_QUALITY_REPORT.md:134:Evaluation uses an owned golden corpus, **grown from n=6 to n=11 this cycle** by adding 5 real
docs\RALLY_DETECTION_QUALITY_REPORT.md:171:re-runs) re-scores the golden corpus under two configs ? **DEFAULT** (served production windowing, all
docs\RALLY_DETECTION_QUALITY_REPORT.md:296:GPU person-feature extraction finished all 6 original golden videos; the LOVO drivers measured the lift
docs\RALLY_DETECTION_QUALITY_REPORT.md:468:scale** ? and this cycle shows that discipline paying off. On an owned golden corpus grown to **n=11**
docs\RALLY_QUALITY_RESEARCH.md:149:- **Corpus today: 15 golden videos** (started at the 6-video label-set hash `50d98ffbc370`, since
docs\RALLY_QUALITY_RESEARCH.md:209:### 4.3 The golden corpus + active learning (the highest-leverage lever)
docs\RALLY_QUALITY_RESEARCH.md:225: golden videos can't. **It is a shadow signal only.** Golden labels remain the gate; Gemini is not
docs\RALLY_QUALITY_RESEARCH.md:231:variant + the golden manifest, emit the full §4.1 metric set + the honest stats block + a leaderboard

```
docs\RALLY_QUALITY_RESEARCH.md:436:  over-merge shows up as a *number* (low Edit, high merge-rate) on the 6 golden videos, and a
docs\RALLY_QUALITY_RESEARCH.md:537:| **E1** | merge/split-rate metric + F1@k surfaced + one-command `rally_seg_eval` entrypoint; **froze baseline** (#185) | 0 | ? | over-merge shows up as a number on the 6 golden videos | ? baseline F1 0.300, merge_rate 0.523 |
docs\README.md:2:[Omitted long matching line]
docs\README.md:18:[Omitted long matching line]
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\lovo_resweep.py:2:golden corpus. Answers: at this n, do the tuned values (cap=6 + R4) still hold, and would
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\README.md:7:The golden corpus grew from n?6 (when cap=6 + R4 were tuned) to **n=13** ? two Boxhill clips just
docs\STORAGE_SHARING_MODEL.md:97:   golden corpus grows ? owned model climbs the quality table (heuristic 0.48 ? Gen-0 0.74 ?
docs\UI_PROPOSAL.md:16:item in NEXT_STEPS is growing the golden corpus, and the v1 scope already commits to a
```

12. assistant (2026-07-03T16:40:22.843Z)

Now let me check for specific tracker/deliverables duplication by examining the existing docs more carefully:

13. user (2026-07-03T16:40:23.717Z)

```
# Golden-labeled video registry

**Purpose.** The single source of truth for every **golden** (human rally-boundary-labeled) video ? what
it is, where the source proxy lives, how many rallies, and whether it's ingested into the eval corpus.
Use this to **isolate the golden videos into a dedicated location** and to keep the growing golden set
auditable. Updated whenever a new video is labeled + ingested.

> **This is the GROWING GOLDEN-finalized set** in the dual-eval-set discipline (see
> [QUALITY_ITERATIONS.md](../QUALITY_ITERATIONS.md), the 2026-06-16 capstone + `eval-dual-set-regression`):
> a **RAW eval set** (currently-bad videos) + this **golden-finalized eval set** (promoted when at-par) +
> a separate **golden TRAINING set**. Every quality launch must not regress on either eval set.
> Privacy-safe, video-free *shipping* of these = [GOLDEN_REGRESSION_FIXTURES.md](../GOLDEN_REGRESSION_FIXTURES.md).

## Contracts / conventions
- **Labels:** rally-annotator CSV (`rally_number,start_time,end_time,ending_reason,sport,shots_count`,
  decimal seconds) ? ingested by `backend/eval/gt_loader.py` (reads `start_time`/`end_time`).
- **Alignment is mandatory before trusting any score.** The label timebase MUST match the windowed proxy
  (same fps + frame count). Verify with `ffprobe` (the `largetest_doubles` near-miss is why) ? ideally
  annotate the *exact* proxy the detector ran on (then `video_id` = MD5 matches and alignment is free).
- **`video_id`** = MD5 of the proxy (`backend/utils/hashing.py::compute_video_id`); it keys the WASB
  trajectory cache, the Gemini reference rows, and the golden features.
- **Ingestion** = copy the trajectory + labels to durable paths, write `output/<name>_golden_features.json`
  (`gts` + candidates), append a row to `output/human_lovo_manifest.json`. (`scratch/at_par.py` scores a
  single video vs Gemini + golden; `backend/eval/rally_seg_eval.py` runs the whole corpus.)
- **Guardrails:** owned footage only; **minors excluded**; labels never derived from a paid LLM (Gemini is
```

```
  inference/reference only). Commercial-clean.

## Ingested golden corpus ? `output/human_lovo_manifest.json` (n = 15)

| # | name | sport | rallies | fps | source proxy | video_id | labeled |
|---|---|---|---|---|---|---|---|
| 1 | adarsh_avi_singles | badminton (S) | 96 | 59.94 | (original 6) | ? | pre-06-16 |
| 2 | kushagra_singles | badminton (S) | 32 | 59.94 | (original 6) | ? | pre-06-16 |
| 3 | largetest_doubles | badminton (D) | 49 | 29.97 | (original 6) | ? | pre-06-16 |
| 4 | gbaaddy | badminton | 48 | 59.94 | (original 6) | ? | pre-06-16 |
| 5 | testlarge_short | badminton | 6 | 29.97 | (original 6) | ? | pre-06-16 |
| 6 | mahadevpura_singles | badminton (S) | 31 | 29.97 | (original 6) | ? | pre-06-16 |
| 7 | mahadevpura_2 | badminton (multicourt) | 40 | 59.94 | `?/Roora Request - June 15/[Done] Video 3 ?/mahadevpura-2_proxy.mp4` | `038e0e0915a2` | 2026-06-16 |
| 8 | GX010128 | badminton (multicourt) | 52 | 59.94 | `?/Golden Label Videos Request - June 15/[Done] Video 13 ?/GX010128.MP4` | `db84f3e65318` | 2026-06-16 |
| 9 | mahadevpura_1 | badminton (S, continuous) | 10 | 59.94 | `?/[Done] Video 14 - Badminton/mahadevpura-1.MP4` | `38bef75d78cb` | 2026-06-17 |

## Labeled, ingestion in progress (WASB trajectory extracting)

| name | sport | rallies | fps | source proxy | status |
|---|---|---|---|---|---|
| GX030094 | badminton | 41 | **29.97** | `?/Golden Label ? /[Done] Video 6 - Badminton/GX030094_proxy.mp4` | **ingested** (in `human_lovo_manifest.json`) |
| Badminton BXH 2 | badminton (multicourt) | 44 | 59.94 | `?/[Done] Video 10 - Badminton Multicourt/Badminton BXH 2_proxy.mp4` | **ingested** (in `human_lovo_manifest.json`) |

## Source folder & not-yet-labeled
- **Annotation root:** `G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15\`
  (+ the earlier `Roora Request - June 15\`). `[Done]` prefix in a dir name = golden-labeled.
- **Labeled badminton dirs:** Video 3 (=mahadevpura-2), 6 (=GX030094), 10 (=BXH 2), 13 (=GX010128, a
  re-label of #8), 14 (=mahadevpura-1). **Video 13 is a duplicate of #8** (already ingested).
- **Not yet labeled / other sports:** Video 1, 4, 7, 9, 12 (badminton, pending); Video 2, 8 (table tennis);
  Video 5, 11 (pickleball). Badminton is the current focus (alpha customers); TT/pickleball labeled later.

## How to isolate the golden videos
Each row's **source proxy** column is the file to copy into a dedicated golden-videos location. The labels
live beside the proxy (`<name>.rallies.csv`) and a durable copy is in `output/<name>.rallies.csv`; the
trajectory is in `Collect/Trajectories/` (path in the manifest). Keep this table in sync on every new ingest.

```

14. user (2026-07-03T16:40:24.087Z)

```
# R2 ? Task-faithful rally-boundary eval metric (tolerance-match + over-seg guard)

**Status:** **IMPLEMENTED (augmenting), 2026-06-17** ? owner-reviewed + decisions locked 2026-06-17;
`backend/eval/tolerance_metrics.py` + `rally_seg_eval` reporting landed (default-additive; tIoU stays the
gate pending the ablation experiment). Gates the rally-end work (R4) and the dual-eval-set regression gate.
A focused increment of the rally-quality project ([RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) §7;
the day-1 validation that motivates it: [REAL_FOOTAGE_VALIDATION_REPORT.md](REAL_FOOTAGE_VALIDATION_REPORT.md);
```

the directive: `eval-boundary-tolerance` + `eval-dual-set-regression`).

> **Implementation refinement (2026-06-17):** §2.3.2's boundary-error distribution is computed over
> **unconditional best-overlap** matches, NOT the within-? 1:1 matches ? a ?-gated set is survivorship-biased
> (loose ends fail the gate and vanish, hiding the very end-deficit the diagnostic exists to surface; that
> within-? deficit instead shows up as low `tol_recall`). Matches the day-1 validation method. Measured on the
> n=9 corpus: **|?start| 1.25 s vs |?end| 2.19 s** ? "start?solved, end=the gap", confirmed.

---

0. TL;DR
`tIoU-F1@0.5` is the wrong ruler for rally detection, in **both** directions: it over-penalizes the
intrinsically fuzzy rally START (the ~1?1.5 s service window), and its overlap matching is murky about
over-production. R2 replaces the headline with a **tolerance-match metric under strict 1:1 (Hungarian)
assignment**, reported as a **decomposition** ? `P@? / R@? / F1@?` + **boundary-error distributions split
into start/end** + an **over-segmentation guard** + a **?-sweep** ? not a single number.

Honest, slightly uncomfortable consequence (measured, see §3): under R2 the local detector's headline
number *drops* and the gap to Gemini *widens* vs what tIoU showed ? because R2 faithfully charges for the
1.5× over-production and the loose ends that tIoU's overlap matching partly absorbed. That is the point:
R2 splits "behind Gemini" into its two real causes ? **(a) over-production [precision]** and **(b) loose
rally-ends [the R4 lever]** ? and forgives only the inherent start fuzziness. It is a more honest ruler,
not a flattering one.

---

1. Why `tIoU@0.5` is the wrong ruler (grounded in the n=2 ground-truth clips)
1. **Over-penalizes the START.** For a 5.7 s rally, `tIoU ? 0.5` tolerates only ~±1.9 s of boundary shift
   (`? = D·(1??)/(1+?)`). The service window (stance ? racquet contact) alone is ~1?1.5 s ? even **Gemini's**
   median |?start| is 0.98?1.46 s. So tIoU spends most of its budget on a boundary that is *intrinsically*
   undefined to ~1 s. (Owner directive: don't strictly align starts; tolerate ~±2 s.)
2. **Count parity is a deceptive proxy.** mahadevpura-2 hit 1.03× of Gemini's count yet F1@0.5 = 0.150 ?
   right count, wrong boundaries. A count- or overlap-based view hides this.
3. **It flips config rankings on noise.** The hard-cap was golden-neutral at n=6 (+0.006 n.s.) but "best" at
   n=7 once a continuously-active clip entered ? a metric artifact at the 0.5 cliff, not a quality change.
4. **The deficits we must see are start?end-symmetric.** Local's |?start| ? Gemini's; the **entire** gap is
   the rally-**END** (|?end| 2.1?3.1 s vs Gemini ~0.8?1.2 s). A single blended IoU cannot show that ? and
   directing R4 needs exactly that decomposition.

---

2. The metric

2.1 Match rule (asymmetric tolerance ? forgive the start, expose the end)
A predicted window matches a golden rally iff **`|?start| ? ?_start` AND `|?end| ? ?_end`**.
- **`?_start` generous (~2.0 s)** ? absorbs the service-window/annotator start fuzziness (the part that is *not* a model deficit).
- **`?_end` tighter (~1.5 s, but IAA-calibrated ? see §2.4)** ? the rally-end IS a sharp physical event (shuttle down/net/out), so a loose end *is* a real deficit we want the metric to expose, not forgive.

> A *symmetric* ±2 s tolerance is tempting (the owner's stated bar) but the preview (§3) shows it is **not**
> automatically more lenient than tIoU 0.5 and it blurs exactly the end-deficit we need to see. Asymmetry is
> the point: relax the dimension that's intrinsically fuzzy (start), keep honest the one that's a real event (end).

2.2 Assignment ? strict 1:1 (Hungarian), never overlap/greedy
Match references to predictions by **global Hungarian assignment** (max total matches under the tolerance gate), one-to-one. This is what makes **over-production cost precision directly**: `P@? = matched / n_pred` crashes when the detector emits 75 windows for 52 rallies. (tIoU's overlap matching let extras "match something"; the W1 over-segmentation pendulum was barely costed.)

2.3 Report the DECOMPOSITION, never one number
1. **`P@?`, `R@?`, `F1@?`** ? precision (over-production), recall (did we find the rally within tolerance), F1.
2. **Boundary-error distributions** ? median **+ IQR** of signed *and* absolute `?start`, `?end` over matched pairs, **split**. This is the metric that says *"start solved, end open"* and aims R4.
3. **Over-seg guard (mandatory):** `split_count` (preds covering one ref ? W1 over-seg) and `merge_count` (refs under one pred ? the baseline mega-window). The *original* rule was strict per-video: **a promotion may not increase either on ANY video.** **REFINED to a NET guard, 2026-06-17 (R1, owner-approved)** ? the strict rule is correct for an *incremental* cue but **mis-fires on a structural mega-window?bounded change** (splitting a mega-window necessarily lifts `split_count` from ~0, and the raw merge *count* is non-monotonic: one window hiding 40 rallies is `count=1, rate=1.0`; bounding it into pieces that each glue 2 neighbours is `count=9, rate=0.5` ? FEWER rallies hidden). The net guard scores over-seg on the **normalized rates** as one weighted cost per video (`w_merge·merge_rate + w_split·split_rate`, merges weighted higher ? a merge hides a rally = recall loss, a split only fragments one), and **passes iff (a) the corpus-mean net cost does not rise AND (b) no video's `merge_rate` regresses.** Implemented as `tolerance_metrics.over_seg_guard` (it always reports the strict verdict too; revert = read `strict_passes`). See QUALITY_ITERATIONS.md "R1" for the measured verdict (milestone: mean `merge_rate` 0.694?0.150, net PASS, strict BLOCK).
4. **?-sweep** (`? ? {1.5, 2.0, 2.5, 3.0}`, symmetric for the trend; asymmetric pair for the headline) ? because the convention/IAA noise is large, a single ? misleads (§3). Report the curve.
5. **`tIoU`/`mIoU` demoted to a SECONDARY trend-line diagnostic** (keeps the paper-ablation backbone comparable) ? **not the gate.**

89 ### 2.4 Choosing ? ? it MUST be calibrated to inter-annotator agreement
90 The preview (§3) shows even **Gemini** scores `tolF1@1.5 = 0.20` on mp2 ? i.e. on ~half the rallies the
91 golden END differs from Gemini's by >1.5 s. That is plausibly **label/convention noise, not model error**:
92 we'd be measuring how two humans (or human-vs-Gemini) disagree on "when did the rally end." **Therefore
93 `?_end` must be ? the inter-annotator end-agreement**, which we do not yet know. **Action:** a small **IAA
94 study** ? a 2nd labeler re-labels 2?3 already-golden videos; set `?_start`/`?_end` to ~the 90th-percentile
95 pairwise human |?start|/|?end|. Until then **freeze provisional ?_start=2.0, ?_end=1.5 and always report the
96 sweep**, with the `? = D·(1??)/(1+?)` derivation recorded so the choice is auditable.
97
98 ---
99
100 ## 3. Preview on the two ground-truth clips (greedy-1:1 approximation; Hungarian in the impl)

15. assistant (2026-07-03T16:40:26.636Z)

Now let me check for any P10/P11 tracker cross-references and content that might duplicate:

16. user (2026-07-03T16:40:27.972Z)

docs\ALPHA_LAUNCH_READINESS.md:35:| D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13.** | sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha can use manual/offline corpus promotion; automated flywheel follows after serving and gates are stable. |
docs\ALPHA_LAUNCH_READINESS.md:82:| P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but can consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha lane is already stable |
docs\ALPHA_LAUNCH_READINESS.md:109:| A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train|reeval` job, and delta-F1 surface | `khelsutra` + `badminton-highlight-indexer` + collector | A1, A3 | Owner D13 reopen | GATED | - |
docs\ALPHA_LAUNCH_READINESS.md:123:1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or should P10/P11 be reopened before any tester wave?
docs\ALPHA_LAUNCH_READINESS.md:139:- [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for alpha-plus.
docs\DOC_STATUS.md:49:| `DECISION_SNAPSHOT_REGRESSION.md` | **DRAFT ? decisions LOCKED** (D1?D11 locked 2026-06-25, #383). Zero code shipped for P1?P7. | No | Start P1+P2 (schema + stitch-plan factor); de-risk the P10 main.py split |
docs\archives\past_projects\CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md:10:> **Status:** `PHASE 1 COMPLETE` + **Phase 2 largely shipped** (P8/P9/P11?P15/P17 merged; remaining = **P10** `main.py` split + Phase-3 P16+) · **Owner:** Avi · **Created:** 2026-06-24 · **Last updated:** 2026-07-02 (P10 currency + §6 anchor fix ? see Changelog)
docs\archives\past_projects\CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md:200:| **P10** | **D1: Split `main.py` god-module** (1448 at audit; **2360 as of 2026-07-02** ? smaller modules) | P7 | No | ? | step-1 extraction [#390](https://github.com/Khelsutra/badminton-highlight-indexer/pull/390) is **stale** (opened 2026-06-26; `main.py` has since grown 1448?2360 across the merged commits) ? re-cut fresh from `origin/master` |
docs\archives\past_projects\CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md:201:| **P11** | **O1: Gemini remote-file GC** ? per-call `_safe_delete` (#373) + startup orphan sweep scoped to our `UPLOAD_TAG`-tagged files, grace-gated, HTTP-timeout-bounded | P7 | No | ? | [#380](https://github.com/Khelsutra/badminton-highlight-indexer/pull/380) |
docs\archives\past_projects\CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md:223:- [x] P5: `create_app()` injects shared state via `app.state` (handlers read `request.app.state`); `backend.api.job_worker` removed from mypy quarantine. (The Optional module globals `db_instance`/`global_config` are **retained as a transitional back-compat shim** and `backend.main` stays mypy-quarantined ? full removal is follow-up, see D1/P10.)
docs\archives\past_projects\CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md:241:| ~~2~~ | ~~**O1 / P11**~~ | ~~Gemini remote-file GC~~ ? **? DONE (#380).** Per-call `_safe_delete` (#373) + startup orphan sweep scoped to `UPLOAD_TAG`-tagged files, grace-gated, HTTP-timeout-bounded. |
docs\archives\past_projects\CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md:249:| 5 |

D1 / P10 | Split `main.py` god-module | **2360 lines as of 2026-07-02** (1448 at audit ? every endpoint/fix PR lands here, so it compounds). D2 (app-factory) was the necessary precursor ? now done. ? The step-1 extraction (**#390**, `backend/orchestration.py`) is stale ? re-cut from `origin/master`. Extract: CLI handlers ? `backend/cli/`, the newer `/api/source`/`/api/assets`/`/api/capabilities` routes ? `backend/api/`, debug helpers ? `scripts/`. |
docs\archives\past_projects\CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md:285:[Omitted long matching line]
docs\archives\past_projects\KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md:80:| `R1-01` main.py line count | Still open, number changed | `backend/main.py` is now **2360 lines** (was ~1965 when this plan was bootstrapped; the Wave-1 endpoint/fix PRs since land here), still large; `backend/orchestration.py` absent; PR `#390` remains open/stale (tracked as `CODE_CLEANUP_AUDIT` P10 ? re-cut fresh from `origin/master`). |
docs\archives\past_projects\KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md:105:**#390 is out of scope** (owner excluded it) and left open; it is tracked as `CODE_CLEANUP_AUDIT` P10
docs\archives\past_projects\KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md:111:in-bucket per §7), and the `CODE_CLEANUP_AUDIT` §6-anchor + P10 currency (`main.py` 1448?2360, #390 stale).
docs\archives\past_projects\KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md:173:| P10 | 3 | Update engine `obsreport` dependency URL to `Khelsutra/` and immutable tag/commit convention. | engine | P0 | No | Todo | ? |
docs\archives\past_projects\KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md:174:| P11 | 3 | Add minimal CI to `sports-obsreport` and fix `_plural("match", 3)` customer text. | sports-obsreport | P0 | No | Todo | ? |
docs\archives\past_projects\KHELSUTRA_GURU_HEALTH_REMEDIATION_PLAN-FOLDED-2026-07-02.md:241:[Omitted long matching line]
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:13:[Omitted long matching line]
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:58:**PR #390 (D1/P10 main.py split) is stale and CONFLICTING while main.py grew 52% since it was cut**
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:59:- anchor: `backend/main.py:1` · tracked in: PR #390; docs/archives/past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md §6 row P10 · finder: engine:audit-doc-reconcile
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:61:- why it matters: This is the last HIGH-severity item of the cleanup audit (P10, the only non-Phase-3 ? row in its §6 tracker) and it is compounding: every new endpoint PR lands in the god-module, widening the conflict surface and making the split more expensive each week. The 3-file extraction in #390 is now unmergeable.
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:64:[Omitted long matching line]
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:86:- anchor: `backend/api/job_worker.py:75` · tracked in: docs/archives/past_projects/CODE_CLEANUP_AUDIT_2026-06-24-SUPERSEDED-2026-07-02.md P5 note + D1/P10 follow-up (line 221) · finder: engine:structure
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:177:[Omitted long matching line]
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:179:- proposed fix: One docs-only reconcile PR: refresh DOC_STATUS.md:47 row (Phase 2 nearly done, remaining = P10 + P16+), add a DECISION_SNAPSHOT_REGRESSION row, fix docs/README.md:41-42 blurbs, and rewrite the NEXT_STEPS.md:6 'Where we are' paragraph to the post-06-21 state.
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:181:[Omitted long matching line]
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:199:[Omitted long matching line]
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:262:[Omitted long matching line]
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:638:| `badminton-highlight-indexer` | `c30ff22` | clean before late-cleanup tracker branch | P0-P10, P15, P23-P25, P27, and final tracker reconciliation landed. |
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:642:| `sports-obsreport` | `a0c7252` / tag `v0.1.3` | clean | P11 landed: CI coverage gate + `_plural("match", 3)` fix. |

docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:672:- Engine `trackers` and `obsreport` dependency pins were mutable/duplicated at audit time; tracker P9/P10 resolved them with immutable commit refs (`#434`, `#435`).
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:688:| #420 | R1-13/14 docs reconcile | Closed ? 4/7 files superseded by #421/#423; 3 residuals carried into #429 | yolo_hybrid docstring + DATA_IN_GCS §5 + CODE_CLEANUP §6/P10. |
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:750:| P10 | 3 | Update engine `obsreport` dependency URL to `Khelsutra/` and immutable tag/commit convention. | engine | P0 | No | Done | #435 |
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:751:| P11 | 3 | Add minimal CI to `sports-obsreport` and fix `_plural("match", 3)` customer text. | sports-obsreport | P0 | No | Done | sports-obsreport #1 |
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:816:- `2026-07-02` ? **P10 obsreport pin (`#435`).** Updated the engine `reporting` extra from the stale `avidullu/sports-obsreport` SSH redirect and mutable `v0.1.3` tag to the canonical HTTPS `Khelsutra/sports-obsreport` repo pinned at commit `a0c7252c6e4159f8702360bea1cb46e5148b5c9f` (sports-obsreport `#1`: CI coverage gate + `match` plural fix).
docs\STORAGE_SHARING_MODEL.md:96:2. **The corpus flywheel becomes the business model:** free users correct rallies (PR-4) ?

17. assistant (2026-07-03T16:40:30.649Z)

Let me do one more targeted check to identify any sections that may state claims as separate sources of truth:

18. user (2026-07-03T16:40:31.737Z)

docs\ABLATION_LAYER.md:59:The corpus is the 6 persisted golden feature JSONs (`output/*_golden_features.json`, written by
docs\ALPHA_LAUNCH_READINESS.md:13:The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has 15 badminton videos; `[analysis]` the local manifest needed path rebasing before manifest-driven commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion schema; `[analysis]` the n=15 quality baseline is stale and must be intentionally refreshed before it can gate anything.
docs\ALPHA_LAUNCH_READINESS.md:33:| D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** | `data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature completeness, baseline refresh, and gate ratification come before quality claims. |
docs\ALPHA_LAUNCH_READINESS.md:37:## 3. Verified snapshot, 2026-07-03
docs\ALPHA_LAUNCH_READINESS.md:68:The quality lane is successful when the 15-video corpus is portable, reproducible, feature-complete, and backed by refreshed baselines. Only then should R2 tolF1, Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.
docs\ALPHA_LAUNCH_READINESS.md:80:| 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still n=6 today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
docs\ALPHA_LAUNCH_READINESS.md:81:| Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly reported stale baselines | Refresh baselines only after reviewer sign-off and record the command/output |
docs\ALPHA_LAUNCH_READINESS.md:101:| A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2 | Reviewer sign-off | TODO | - |
docs\ALPHA_LAUNCH_READINESS.md:104:| A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data availability | TODO | - |
docs\ALPHA_LAUNCH_READINESS.md:117:4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor refresh.
docs\ALPHA_LAUNCH_READINESS.md:127:5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every clip, or should the eval loaders support mixed schemas with explicit capability tags?
docs\ALPHA_LAUNCH_READINESS.md:134:- [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
docs\CODE_MAP.md:7:[Omitted long matching line]
docs\CODE_MAP.md:124:[Omitted long matching line]
docs\CODE_MAP.md:175:- `@backend/eval/classifier.py::LogisticRegression` ? numpy-only logreg (no

sklearn). `fit/predict_proba/predict/to_dict/from_dict/save/load`. ? `class_weight='balanced'` default; stores feature standardization (mean/std) ? inference MUST use same feature ORDER; `_sigmoid` clips z to [-30,30]; `lr=0.1,epochs=1000,l2=1e-3`; JSON `"model":"logreg"` tag.
docs\CODE_MAP.md:336:| Guard rally-quality nightly (no GPU) | `@scripts/nightly_regression.py::main` ? re-score golden corpus (default + milestone configs) vs `eval_baselines/nightly_baseline.json`; exit 1=regress, 4=stale; `--update-baseline` to re-snapshot after an intended change. Scheduled via `scripts/register_nightly_task.ps1` |
docs\CODE_MAP.md:337:| Guard the FULL pipeline + reel-count nightly (GPU) | `@scripts/nightly_reelcount.py::main` ? run the real configured+checkpointed pipeline on gs:// clips, assert reel-count EXACT + quality + report cost, email it; exit 1=regress, 4=stale; `--update-baseline` to re-freeze. Runs on an ephemeral GCP GPU VM (`deploy/gcp/nightly_regression/launch.sh`) that self-deletes |
docs\ANALYZERS_AND_RECONCILERS.md:291:feature_names=[...])` (`experiment.py:543`) reads each name out of the row's `stats` JSON into a feature
docs\EXPERIMENT_HARNESS.md:86:`backend/eval/classifier.py` ? `LogisticRegression` (pure numpy, L2): `fit(X,y,feature_names=,class_weight="balanced")` (:40), `predict_proba` (:75), `predict(threshold=0.5)` (:81), `to_dict`/`from_dict` (:86/:97), `save`/`load` (JSON) (:107/:111). **A trained model = one JSON file ? a pinnable, hashable variant artifact.**
docs\DECISION_SNAPSHOT_REGRESSION.md:34:| **D5** | **Snapshot only deterministic segmenters/fields; exclude the random fabricated fields.** Use a deterministic segmenter (CONTROL/fusion on trajectory) and a `{start,end,shuttle}`-style schema that excludes `motion`'s random `server/receiver/shot_count/avg_speed`. | `motion` fabricates random fields (audit **B1**) [verified] ? non-snapshot-able as-is. **B1's fix (drop the fabrication) is the enabler that would make `motion` snapshot-able.** | **LOCKED** (2026-06-25) |
docs\NEXT_STEPS.md:61:3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir <shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir <shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
docs\MULTI_SIGNAL_FUSION_PLAN.md:143:These five person features are now **built** (`fusion_features.py::PERSON_FEATURE_NAMES`; default-OFF, unmeasured on golden) and join the shuttle features in the candidate `stats` JSON
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:45:[Omitted long matching line]
docs\RALLY_DETECTION_QUALITY_REPORT.md:455: configs; exit 1 = regression, 4 = stale baseline.
docs\README.md:18:[Omitted long matching line]
docs\archives\past_projects\DECOUPLED_COMPUTE\07-COMPUTE-ABSTRACTION.md:58:`Frame,Visibility,X,Y` CSV or a feature JSON). Everything downstream is device-agnostic CPU that reads that
docs\RALLY_QUALITY_RESEARCH.md:147: `fusion_compare.py` / `fusion_audio.py` ? extract replayable per-window feature JSONs
docs\data_pipeline\GOLDEN_DATA_SHARING.md:42:   ? real person-feature ?F1 / CI / sign test. Copy the JSONs into the repo's `output/` to also light up the `skipif` golden-litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`).
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:181:[Omitted long matching line]
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:190:[Omitted long matching line]
docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:289:[Omitted long matching line]

19. assistant (2026-07-03T16:40:49.812Z)

Based on my thorough review of the PR #451 diff and the existing documentation, I've identified the key duplication and drift risks. Let me compile my findings:

```json
[
  {
    "file": "docs/ALPHA_LAUNCH_READINESS.md",
    "line": 51,
    "summary": "Section 3 'Verified snapshot, 2026-07-03' restates n=15 golden corpus facts from GOLDEN_VIDEOS.md without pointing to it as the source of truth. The manifest path, video count, and feature completeness details are now stated in two places.",
    "failure_scenario": "GOLDEN_VIDEOS.md §1 is the documented single source of truth for 'every golden video ? what it is, where...how many rallies...whether it's ingested'; the new doc
```

restates corpus facts independently, creating two maintenance points that will diverge when new videos are ingested (only one doc will be updated)."
  },
  {
    "file": "docs/ALPHA_LAUNCH_READINESS.md",
    "line": 81,
    "summary": "Section 5 'Risks and guardrails' states 'Stale local manifests create false confidence' and proposes A2 'portable manifest tooling' as the fix. This duplicates the baseline refresh and corpus-portability tracking already live in NIGHTLY_ENGINE_REGRESSION.md and the baseline-update convention.",
    "failure_scenario": "NIGHTLY_ENGINE_REGRESSION.md already tracks baseline staleness as a primary gate; the new doc introduces a parallel 'stale baseline' concern and separate refresh/recording ceremony in §7 A4, creating two competing trackers for when and how baselines are refreshed."
  },
  {
    "file": "docs/ALPHA_LAUNCH_READINESS.md",
    "line": 103,
    "summary": "Row A7 'Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15' restates the feature-completeness problem noted in §3 snapshot. The fusion feature gap (only 6 of 15 have full schema) is a measurement fact, not an action, and belongs with corpus registry, not tracked separately as a deliverable.",
    "failure_scenario": "GOLDEN_VIDEOS.md line 47?48 will track when each video is 'ingested' (features extracted); this doc independently tracks feature regeneration as deliverable A7, creating two separate sources of truth for 'which videos have complete features' that will drift."
  },
  {
    "file": "docs/ALPHA_LAUNCH_READINESS.md",
    "line": 109,
    "summary": "Row A12 'Reopen or defer P10/P11 corpus flywheel' treats P10/P11 as an alpha-launch decision with a gated tracker row. However, COMPUTE_DECOUPLED_SERVING/README.md §7 M11 already owns the 'data-flywheel harness' delivery and its dependencies/gating (M7,M9,owned-model/golden data).",
    "failure_scenario": "Two docs now separately track the same flywheel work (M11 vs A12) with different gating and sequencing. If the owner reopens P10/P11 or changes the owned-model timeline, only one of these trackers will be updated, causing confusion about which is the active tracker."
  },
  {
    "file": "docs/ALPHA_LAUNCH_READINESS.md",
    "line": 13,
    "summary": "The TL;DR snapshot of 'only 6 of 15 feature JSONs had the full fusion schema' and 'the n=15 quality baseline is stale' are analysis findings from 2026-07-03 local runs. These are restated in §3 as verified facts but lack a link to the authoritative source (GOLDEN_VIDEOS.md for corpus facts, NIGHTLY_ENGINE_REGRESSION.md for baseline state).",
    "failure_scenario": "The snapshot claims are local-verified at a point in time, but they don't reference the living docs (GOLDEN_VIDEOS.md, baseline tracker) where the current truth lives. Future readers will treat the snapshot as current fact, not a historical record, and won't know to update the baseline tracker instead."
  },
  {
    "file": "docs/ALPHA_LAUNCH_READINESS.md",
    "line": 91,
    "summary": "Section 7 deliverables tracker (rows A0?A13) creates a NEW cross-repo execution tracker without consolidating into or explicitly deferring the existing per-doc trackers: RALLY_DETECTION_FIXES_PLAN.md §7, R2_EVAL_METRIC_DESIGN.md (ablation as a gate), and COMPUTE_DECOUPLED_SERVING/README.md §7 (M11 data-flywheel). Some rows (A6 ablation, A7 fusion features, A12 flywheel) overlap existing tracked deliverables.",
    "failure_scenario": "Row A6 'Fix ablation CLI circular import + run R2/tolF1 ablation on n=15' is a new entry for an existing gate (R2 ablation, already in R2_EVAL_METRIC_DESIGN.md as the gating condition). If the R2 ablation lands and is recorded elsewhere, this tracker won't be updated, and future readers won't know which PR represents the completion of 'the R2/tolF1 ablation on n=15'."
  }

]
```
```